MpGI: Multiperspective Generation and Integration for High-Quality Remote Sensing Image–Text Datasets

Yiguo He, Junjie Zhu, Jun Wang, Yiying Li, Xiaoyu Zhang, Xichuan Zhang, Chunping Qiu, Qiangjuan Huang, and Ke Yang

***Abstract*—Recent advances in remote sensing (RS) vision-language foundation models (VLFMs) rely heavily on large-scale paired image–text data. However, existing dataset construction methods mainly emphasize data scale while overlooking a more fundamental limitation: the lack of structured and comprehensive semantic representation. Since RS imagery inherently contains multilevel and multiperspective semantics, including object attributes, spatial layouts, scene context, and functional relationships, single-pass caption generation often produces incomplete, biased, or weakly aligned descriptions. To address this issue, we propose a *multiperspective semantic distillation paradigm* for constructing high-quality RS image–text datasets. Based on this paradigm, we introduce a two-stage framework termed multiperspective generation and integration (MpGI). In the first stage, we perform *annotation-guided knowledge elicitation* using multiple multimodal large language models (MLLMs), where structured annotations are transformed into model-understandable instructions to extract diverse and complementary visual-semantic descriptions. In the second stage, we conduct *semantic distillation and alignment-aware compression* using large language models (LLMs), which integrate complementary descriptions and reorganize the resulting semantics into compact captions better suited for vision-language learning. With this framework, we construct *HQRS-IT-210K*, a high-quality RS image–text dataset containing approximately 210K images and 1.26M image–text pairs. We fine-tune CLIP to obtain HQRS-CLIP, which outperforms previous state-of-the-art RS vision-language models across multiple downstream tasks, including zero-shot classification (ZSC), few-shot classification (FSC), cross-modal retrieval, and semantic localization (SeLo), while *using only 4.2% of the training data used by previous approaches*. We further fine-tune CoCa on the full dataset to obtain RS-CoCa, which demonstrates strong RS image captioning (RSIC) capability and can generate captions comparable to manual annotations. These results show that improving semantic quality and image–text alignment is a more effective direction than merely increasing data volume for advancing RS vision-language modeling. The dataset, pretrained models, and code are available at https://github.com/YiguoHe/HQRS-210K-and-HQRS-CLIP**

***Index Terms*—CLIP, CoCa, image–text paired dataset, remote sensing (RS), vision language foundation model (VLFM).**

I. Introduction

VISION-LANGUAGE foundation models (VLFMs), such as CLIP [1] and CoCa [2], have demonstrated remarkable capabilities in bridging visual and textual modalities through contrastive and generative objectives on large-scale image–text pairs. Recently, the remote sensing (RS) community has recognized the power of VLFMs and has begun exploring their applications in the field of RS [3], [4], [5], [6], [7], [8], [9], [10]. These studies have demonstrated that VLFMs trained with extensive paired RS image–text data perform well in various RS applications. *The key to achieving success in this area lies in the construction of high-quality, large-scale RS image–text paired datasets.*

High quality refers to two key aspects. For images, the dataset should primarily consist of clear aerial or satellite images while minimizing the inclusion of noisy or irrelevant images [11]. Additionally, images of the same category or scene should be as nonredundant and diverse as possible, ensuring high variability within and across categories [12], [13]. For text, it is crucial to maintain a strong correlation between the textual descriptions and the image content. Furthermore, the text should provide rich and detailed descriptions of the image content [14], [15], allowing for a high degree of alignment between images and text. Accurate and comprehensive descriptions are crucial, as captions that focus on only a small part of an image can cause category ambiguity [16], [17].

However, achieving these goals is highly challenging. Unlike natural images, RS images and their associated textual descriptions cannot be effectively sourced from the public internet. Additionally, manually annotating aerial images requires specialized knowledge and is extremely time-consuming [12]. This challenge is even greater for textual caption annotation, as RS images often lack detailed content, making it difficult even for experts to provide diverse annotations.

To address these challenges, prior work has explored three main directions. Rule-based methods [3], [4] convert metadata or annotations into captions, yet these captions are typically broad, partial, and linguistically rigid due to fixed generation rules. Learning-based approaches, such as RS5M [5], fine-tune captioning models (e.g., BLIP-2) on existing small-scale RS caption datasets; unfortunately, they inherit the original datasets' limitations in brevity, coverage, and diversity. More recent large language model (LLM)-based methods [18]

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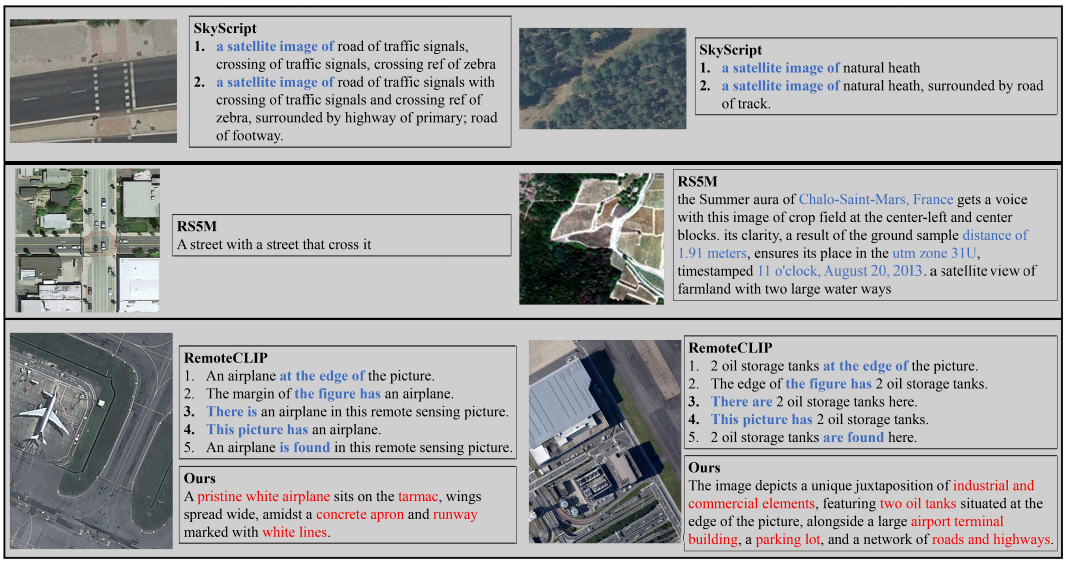


Fig. 1. Samples of large-scale RS image–text datasets. SkyScript captions are rule-based, resulting in highly uniform sentence structures and poor alignment with images. In RS5M, BLIP2 captions are short and lack richness, while metadata-based captions are lengthy but misaligned with visuals. RemoteCLIP captions are repetitive and lack diversity. In contrast, our dataset provides accurate, comprehensive, and diverse captions, rivaling human annotations.

generate longer descriptions from coarse annotations (e.g., land-cover types), but the resulting captions often lack strong visual grounding.

As shown in Fig. 1, existing methods mainly face two issues: insufficient semantic richness and limited linguistic diversity. On the one hand, they often fail to provide semantically rich and comprehensive descriptions for RS images, resulting in inadequate semantic coverage. On the other hand, they tend to generate repetitive, rigid, or weakly grounded language, which limits both linguistic diversity and image–text alignment. We argue that these deficiencies originate from a more fundamental limitation: the lack of **structured and comprehensive semantic representation** in RS image–text pairs. RS images inherently involve multilevel and multiperspective semantics, including object-level attributes, spatial layouts, scene context, and functional relationships. However, most existing approaches rely on a **single-pass caption generation paradigm**, which compresses these heterogeneous semantic aspects into a single generation process. Consequently, the resulting captions are often incomplete, biased, or weakly aligned with the visual content, even when the dataset scale is large. *These observations indicate that scaling data volume alone is insufficient and that semantic quality and image–text alignment deserve more explicit attention in RS vision-language learning.*

To address these limitations, we propose a **multiperspective semantic distillation paradigm** for RS image–text dataset construction. The core idea is to explicitly model RS semantics as a composition of multiple complementary views and then distill these heterogeneous semantic representations into compact, aligned textual descriptions suitable for vision-language learning. Based on this paradigm, we introduce a two-stage framework named multiperspective generation and integration (MpGI), as illustrated in Fig. 2.

In the first stage, we perform **annotation-guided knowledge elicitation** from multiple multimodal LLMs (MLLMs).

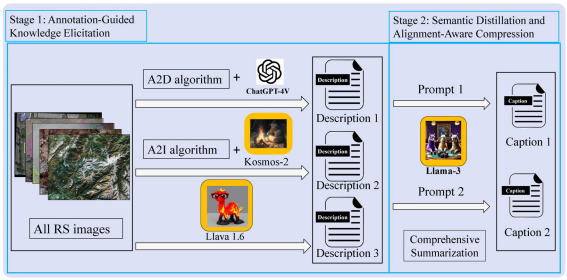


Fig. 2. Overview of MpGI, a two-stage framework for multiperspective semantic distillation in RS image–text dataset construction. Stage 1 elicits complementary descriptions from multiple perspectives under annotation guidance, and Stage 2 distills them into compact, aligned captions for vision-language learning.

Specifically, structured annotations such as class labels, bounding boxes, and segmentation masks are converted into model-understandable instructions, which guide MLLMs to generate diverse and complementary descriptions for each image. This process can be viewed as a form of **data-level knowledge extraction and distillation**, where latent visual-semantic knowledge embedded in large models is externalized under structured supervision. At this stage, we generate detailed descriptions for each RS image with an average length exceeding 220 words.

In the second stage, we conduct **semantic distillation and alignment-aware compression** using LLMs. Given multiple complementary descriptions that capture different semantic aspects of the same RS image, the LLM integrates visually grounded information, mitigates partial redundancy and occasional generation errors, and reorganizes the resulting semantics into a compact representation. This process goes beyond conventional summarization by performing semantic consolidation, error suppression, and compression under an information bottleneck, thereby producing captions that are

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both information-rich and well aligned with vision-language models such as CLIP.

To address the uniform style tendency of LLMs under a single prompt [19], we adopted multiple strategies to enhance caption diversity. These include prompting LLMs to randomly sample outputs and using distinct prompt designs to diversify the results.

Furthermore, we explored a probability-based fusion method, which allowed the final caption set to include various caption styles generated in the second stage. This can be regarded as a text augmentation strategy without increasing the training budget because it enhanced the diversity of the training data. Experimental results show that this simple strategy substantially improved model performance without increasing training costs.

Finally, we developed the HQRS-IT-210K dataset (High-Quality RS Image–Text dataset with 210K images), containing approximately 210 000 RS images and 1.26 million image–text pairs. We used only 1/6 of the data in this dataset to fine-tune CLIP (Specifically, this one-sixth subset contains one sampled caption for each image, forming a one-to-one image–text training set for HQRS-CLIP.), resulting in our HQRS-CLIP models. Experimental results suggest that our HQRS-CLIP outperformed the previous state-of-the-art methods across various downstream tasks, including zero-shot classification (ZSC), few-shot classification (FSC), RS image–text retrieval, and semantic localization (SeLo), *all while using only 4.2% of the training data* ([5]). Then, to validate the effectiveness of our dataset for generation tasks, we fine-tuned the CoCa model using the entire dataset, resulting in RS-CoCa. Experimental results demonstrate that RS-CoCa exhibits significantly improved RS image captioning (RSIC) capabilities, producing captions that can even surpass human annotations in quality.

Our contributions can be summarized as follows.

1) We propose a **multiperspective semantic distillation paradigm** for RS image–text dataset construction, which shifts the focus from single-view generation to structured semantic modeling and alignment. Based on this paradigm, we introduce MpGI, a two-stage framework that generates high-quality RS image–text pairs by combining annotation-guided knowledge elicitation and semantic distillation. Following this framework, we construct the HQRS-IT-210K dataset with approximately 210K RS images and 1.26 million image–text pairs, which has been released to advance vision-language research in RS.
2) With the HQRS-IT-210K dataset, we fine-tune powerful RS VLFMs for both discriminative and generative vision-language tasks. HQRS-CLIP learns robust vision-language representations for diverse RS applications, while RS-CoCa generates captions comparable to or even better than manual annotations.
3) We conduct extensive experiments to analyze the effectiveness of the proposed two-stage framework and show that improving semantic quality enables substantially better performance with much less training data.
4) To better evaluate the long-text understanding ability of RS vision-language models, we further introduce the first benchmark dataset and baselines for long-text image retrieval in RS.

II. Related Work

A. RS Image–Text Datasets Construction

Manual Annotation: Annotating aerial images requires specialized knowledge and is extremely time-consuming [12]. Datasets such as UCMcaption (with 2100 images), Sydney Captions (with 613 images) [20], RSICD (with 10 921 images) [17], RSITMD (with 4743 images) [21], and NWPU-Captions (with 31 500 images) [22] rely on manual labeling. While the captions are accurate, the process is costly and inefficient, which limits the scalability of these datasets.

Public Natural Image–Text Datasets Filtering: Unlike natural images, RS image–text data are scarce online. Czerkawski and Francis [23] used CLIP to extract 28 572 RS images from LAION-EO, but their utility remains unproven. Zhang et al. [5] filtered 3M RS image–text pairs from 11 datasets, but the resulting data still suffered from considerable noise.

Rule-Based Generation Methods: Rule-based approaches generate RS captions by assembling existing annotations or metadata from RS images [3], [4], [5]. However, these captions often suffer from high repetition, lack natural semantics, and contain irrelevant metadata noise that is difficult to align with visual features in the images.

Generative Large Models-Based Generation Methods: Recently, researchers have begun using LLMs and MLLMs to generate RS captions. Yuan et al. [18] employed ChatGPT-3.5 and ChatGPT-4V for RS captioning, but the generated results were limited by simple prompts and insufficient visual grounding. Zhang et al. [5] fine-tuned BLIP-2 on RSITMD to caption millions of RS images. However, because the original RS caption datasets are typically brief and limited in semantic coverage, the resulting model inevitably inherits these deficiencies. As a result, current methods do not fully exploit the capabilities of LLMs and MLLMs for constructing semantically rich and well-aligned RS image–text data.

Discussion: As shown in Table I, existing RS image–text dataset construction methods mainly rely on public RS datasets and increasingly incorporate large generative models. This trend is both practical and effective, since public datasets provide reliable annotations and large models can transform such annotations into more natural language descriptions. However, most existing methods still formulate RS caption construction as a *single-pass generation problem*. They do not explicitly model the inherently multilevel and multiperspective semantics of RS imagery, such as object attributes, spatial layouts, scene context, and functional relations, nor do they explicitly integrate and distill complementary semantics from multiple sources. In contrast, our work treats RS image–text construction as a problem of **multiperspective semantic elicitation and distillation**, aiming to produce captions that are not only natural but also semantically comprehensive and better aligned with vision-language learning.

B. RS VLFMs

In natural language processing (NLP), LLMs like BERT [24], T5 [25], BART [26], and the GPT series [27], [28],

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TABLE I

COMPARISON OF DIFFERENT LARGE-SCALE RS IMAGE–TEXT DATASETS. **IT PAIRS** REFERS TO IMAGE–TEXT PAIRS. **METHOD** REFERS TO THE CAPTION GENERATION METHOD. **DATASET COMMENT** SECTION ANALYZES THE ADVANTAGES AND LIMITATIONS OF EACH DATASET. **R-B** REFERS TO RULE-BASED GENERATION METHOD. **PNDF** REFERS TO THE PUBLIC NATURAL IMAGE–TEXT DATASETS FILTERING METHOD

Model	IT Pairs	Method	Dataset Comment
SkyCLIP	2.6Million	R-B	This approach introduces a large number of new RS images, making a significant contribution. The image labels encompass a wide range of semantic tags, which can enhance the model's generalization ability. **However**, the caption structure is too rigid and repetitive, lacking both natural expression and diversity.
RemoteCLIP	828 K	R-B	By using annotations from public RS datasets, the caption information is accurate. **However**, the sentence structure of captions is rigid and repetitive, lacking natural expression and diversity. Additionally, the captions often describe only a portion of the image, leading to category ambiguity. Furthermore, the captions lack image background information and scene context beyond the annotations.
GeoRSCLIP	5 Million	R-B PNDF MLLM-B	Using fine-tuned BLIP-2 combined with annotated RS labels, it generates captions with natural and varied structures. The dataset includes a large and diverse collection of images, enhancing the model's generalization ability. **However**, the images obtained by PNDF often include irrelevant or suboptimal ones, and many captions contain metadata unrelated to the visual content, which can distract the model from focusing on the visual information. Additionally, BLIP-2 can only utilize class labels when generating captions, resulting in brief descriptions that lack detailed information about the images.
Ours	1.3Million	MpGI	The MpGI method is composed of Rule-MLLM Relay generation, MLLM-Based, and LLM-Based methods. Each image's description is sourced from multiple origins, making it comprehensive and precise. By compressing the Visual information to captions within 77 tokens, the caption information can be fully used by CLIP. Diverse captions capture a wide range of semantic and visual information. **However**, although keyword filtering and word-length filtering were applied to reduce noise, inaccuracies caused by hallucinations during the generation process of MLLMs and LLMs cannot be entirely eliminated.

[29], [30] achieved significant success. Decoder-only models like GPT-3 are pretrained on next-token prediction tasks, demonstrating strong zero-shot and few-shot generalization. Recently, the application of VLFMs to RS imagery has garnered significant attention.

RS VLFMs: CLIP [1] employed a contrastive learning approach, training CLIP on 400 million image–text pairs. Although CLIP has shown impressive performance, it underperforms in specialized fields such as medicine and RS [1], [5]. In these contexts, developing domain-specific CLIP models can offer substantial advantages. Researchers in RS are developing specialized foundational models based on CLIP, such as RemoteCLIP [3], SkyCLIP [4], and GeoRSCLIP [5].

RS MLLMs: Integrating CLIP's image encoders with LLMs yields MLLMs (e.g., BLIP [31], [32], Kosmos [33], [34], LLaVA [35], [36], [37], and Qwen-VL [38]). In RS, BITA [39] builds upon pretrained large models by further training them on RS datasets to bridge the image–text modality gap. RSGPT [9] and RS-LLaVA [8] were developed by fine-tuning InstructBLIP and LLaVA on limited datasets like RSICap and RS-instructions, which involve complex data construction methods and remain small in scale. The SkyEye-968k dataset [10], constructed by integrating existing captioned datasets and manually verified samples, includes 968K samples but heavily relies on existing captioned data, making it costly and less diverse. In contrast to prior RS VLFM and RS MLLM studies that mainly depend on existing captions or single-pass caption generation, our method focuses on **annotation-guided semantic elicitation**, **multiperspective semantic integration**, and **alignment-aware semantic distillation**. This makes the constructed data particularly suitable for training RS VLFMs such as CLIP and CoCa.

III. HIGH-QUALITY IMAGE–TEXT PAIR GENERATION

We collected 210 556 RS images from 23 public satellite and uncrewed aerial vehicle (UAV) imagery datasets and captioned them using a two-stage framework, as shown in Fig. 2.

A. Data Preparation

As shown in Fig. 3(a), we collected 23 datasets and thoroughly cleaned the images and annotations before generating captions. Table II provides a comprehensive overview of the sources of the HQRS-IT-210K dataset, detailing the information for each subdataset.

In terms of images, we first removed unannotated images from each dataset. For images that were too large, we employed a sliding-window approach to partition them into several nonoverlapping smaller image patches. This is to capture the fine-grained details of these large RS images when generating captions, while also serving as a data augmentation strategy. A strict deduplication method using p-hash and URLs was employed to prevent data leakage [3]. Ultimately, about 1/5 of the images were removed. In terms of annotations, denoising was also performed. For example, we removed annotations labeled as "ignored regions" and "others" in the VisDrone dataset. The wording of the original annotations was adjusted where necessary. For example, we changed the annotation for people from "Human" to "person."

B. Stage 1 (Annotation-Guided Knowledge Elicitation)

As shown in Fig. 3(b) and (c), the goal of this stage is to perform annotation-guided knowledge elicitation for each RS image through multiple complementary methods. Specifically, structured annotations are leveraged to guide different MLLMs to externalize diverse visual-semantic knowledge from different perspectives. Each method has its own strengths, and the resulting descriptions provide complementary and enriched information for the same image when combined.

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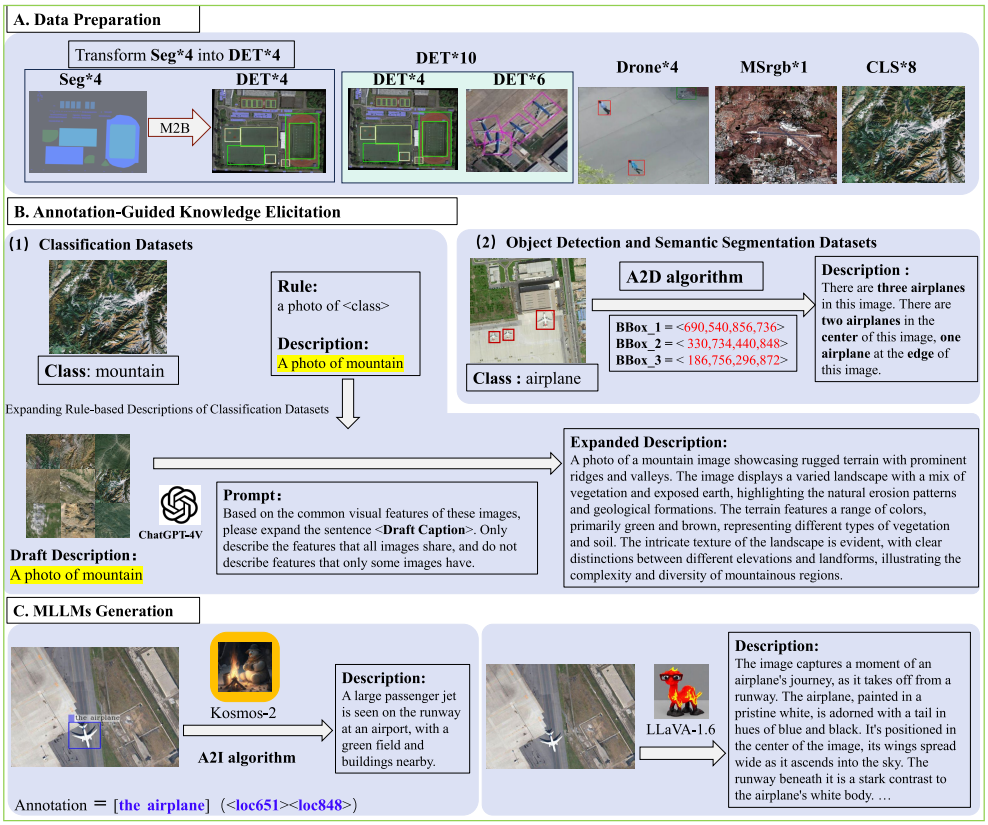


Fig. 3. Pipeline of data preparation and Stage 1 annotation-guided knowledge elicitation. (a) Different RS datasets are preprocessed, transformed, and unified into several data sources. (b) Structured annotations are converted into descriptive texts through rule-based methods. (c) Multiple MLLMs are guided by annotation-derived instructions to generate complementary visual-semantic descriptions from different perspectives.

1) Rule-MLLM Relay Generation: Our images are from classification, object detection, and segmentation datasets. They have been precisely annotated with accurate class labels, bounding boxes, and segmentation masks. Our goal is to efficiently use these precise annotations.

For segmentation masks, we use the mask-to-box (M2B) algorithm [3] to extract the coordinates (xmin, ymin, xmax, ymax) of objects annotated in Seg*4 and convert the segmentation masks into bounding boxes, converting them into DET*4. DET*4 and DET*6 were combined into DET-10 for unified caption generation. As shown in Fig. 3(b-2), for DET-10, we use the annotation-to-description (A2D) algorithm to convert the class labels, object counts, and location annotations into English sentences. When describing the positions of objects, we define the central area as the rectangular region spanning from 1/4 to 3/4 of the image's width and height, with the remaining region defined as the edge area. The A2D algorithm improves upon the B2C algorithm [3] by reducing category ambiguity.

The A2D algorithm includes the following rules.

1) *Rule 1:* Describe all objects annotated in the image. *Example: There are three cars and two trucks in this image.*
2) *Rule 2:* Describe objects located both in the center and at the edge of this image. *Example: There are three cars in the center of this image and two trucks at the edge of this image.*

Algorithm 1 A2D

```
Function Annotation_to_Description(bbox_list):
    descriptions ← [ ];
    // Rule 1: Describe objects annotated in the
       image.
    class_counts ← count_categories(bbox_list);
    description_parts ← generate_description_parts(class_counts);
    descriptions.append("There are " + join(description_parts, ", ") + ".");
    // Rule 2: Describe objects located both in the
       center and at the edges of the image.
    center_counts, edge_counts ← count_center_and_edge(bbox_list);
    center_parts ← generate_description_parts(center_counts);
    edge_parts ← generate_description_parts(edge_counts);
    descriptions.append("There are " + join(center_parts, ", ") + " in the
      center, and " + join(edge_parts, ", ") + " at the edges of the image.");
```

The pseudo-code for the A2D algorithm is shown in Algorithm 1. These rules ensure that all annotated objects are included in the captions, alleviate the risk of category ambiguity [16], and provide a more complete description.

As shown in Fig. 3(b-1), for classification datasets, we start by generating the original descriptive sentence with "a photograph of [class]." Then we use ChatGPT-4V [30] to expand this sentence based on the common features of images in the same class folder. We chose ChatGPT-4V for its exceptional performance. Since each image category requires only one expansion, the cost remains low despite ChatGPT-4V being closed-source. Specifically, for each class, we randomly select 9 images from the class folder and pass them to ChatGPT-4V, which identifies their common features

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TABLE II

OVERVIEW OF HQRS-IT-210K DATASET. FOR THE FMOW DATASET, WE ONLY SELECTED SAMPLES FROM THE VALIDATION SET, AS GEORSCLIP [5] HAD ALREADY GENERATED CAPTIONS FOR THE TRAINING SET, THEREBY AVOIDING REDUNDANCY

Collection	Datasets	Images	Class	Description
MSrgb*1	fMoW	22723	62	The fMoW dataset includes class and bounding box annotations with pan-sharpened RGB images derived from multispectral data to improve resolution and visual detail.
CLS*8	AID	5465	30	In RS classification datasets, images within the same category exhibit high similarity and significant homogeneity. As a result, during the P-Hash deduplication process, many samples were excluded from the original dataset. These RS scene classification datasets only contain class labels, so during rule-based caption generation, GPT-4 is also used to expand the sentences for each category's caption by identifying common features across multiple images of the same category within each dataset.
	RESISC45	13630	45	
	OPTIMAL	1596	31	
	RSI-CB256	9582	35	
	RSI-CB128	13184	45	
	EuroSAT	5101	10	
	WHURS19	846	19	
	MLRSNet	22975	46	
UAV*4	VisDrone	27303	11	To increase data diversity, we introduced Drone aerial imagery datasets, which include detailed class and bounding box annotations. The ablation experiment results indicate that the image-text data constructed from these datasets can enhance the recognition capability of the CLIP model in the RS domain.
	Stanford	3067	6	
	AUAIR	2175	8	
	CARPK	513	1	
Seg*4	iSAID	20786	15	These datasets provide more comprehensive visual information, such as the categories of all objects in an image, thereby offering annotations and textual information with higher alignment accuracy. To ensure consistent processing, we convert segmentation masks to bbox format and process them together with the object detection dataset in subsequent stages.
	LoveDA	4187	6	
	Vaihingen	33	5	
	Potsdam	4515	5	
DET*6	HRRSD	21761	13	These RS object detection datasets provide precise class and bounding box annotations, allowing rule-based generation methods to produce relatively informative captions. However, since these datasets are designed for object detection tasks, many elements that are not emphasized in object detection, such as image scenes, remain unannotated. Nonetheless, textual descriptions of these image features are crucial for image-text alignment tasks. The extraction of such information requires the use of MLLMs.
	LEVIR	3791	3	
	RSOD	936	4	
	DOTA	1869	15	
	DIOR	23463	20	
	HRSC	1055	31	

to expand the original descriptive sentence. We refer to this approach (first generating simple captions using rule-based methods and then expanding them with ChatGPT—4V) as the "Rule-MLLM Relay Generation method," abbreviated as "R-M Relay." Experimental results demonstrate the effectiveness of these strategies.

To address the hallucination issue, we manually conducted a sentence-by-sentence review of all captions generated by ChatGPT-4V to ensure the accuracy of the caption content.

2) Instruction-Guided MLLM Generation: The descriptions generated for DET-10 by the rule-based method lack contextual information beyond annotations, such as details about aerial scenes. The descriptions of the classification datasets lack image-specific details. Therefore, we leverage MLLMs to generate more detailed and specific descriptions for images.

First, we selected the Kosmos-2 [34] model to generate fine-grained information. The selection was motivated by three key reasons: 1) Kosmos-2 is one of the most advanced open-source VLFMs and achieves state-of-the-art performance in image captioning tasks; 2) Kosmos-2 is equipped with visual grounding capabilities, enabling it to link generated textual entities to corresponding image regions via bounding-box tokens; and 3) it has the ability to accept object detection boxes as inputs, allowing us to leverage the bounding box annotations in our DET-10 dataset to generate spatially-aware captions.

Then, we also chose LLaVA-1.6 [37] (also called LLaVA-NeXT) to generate detailed descriptions for each image. First, LLaVA-1.6 and Kosmos-2 differ in model architecture and training data, leading to variations in their captioning abilities. These two models can generate descriptions from different perspectives for the same image, often providing complementary information. Second, the image resolution of Kosmos-2 is set to 224×224, which leads to the loss of contextual details in many large RS images. So we need another MLLM capable of handling high-resolution images to generate detailed descriptions for each image. LLaVA-1.6 supports resolutions up to 672×672, 336×1344, and 1344×336 across 3 aspect ratios, enabling the capture of more visual details. Moreover, LLaVA-1.6 demonstrates exceptional performance, outperforming other open-source MLLMs such as InstructBLIP [32], CogVLM [40], and Yi-VL [41].

The instruction format and prompt have a significant impact on the performance of MLLMs. To achieve better results, we explored the impact of different instructions for Kosmos-2 and LLaVA-1.6 before formal experiments. We randomly selected 1000 images from the overall dataset to test the impact of different instructions.

For LLaVA-1.6, we primarily tested its ability to generate detailed descriptions using instructions with or without class labels. Experimental results showed that the instruction *Describe this image in detail* can guide LLaVA-1.6 to generate accurate and detailed information for RS images.

For Kosmos-2, after careful design and extensive experiments with 10 different instruction templates, we adopted the following 3 instruction templates.

① Describe this image with [class] in detail.
② Describe this image with [class + bbox] in detail.
③ Where is/are the [class + bbox]? Answer.

For images with only a class label, we use ① to generate accurate and comprehensive descriptions. For images with one or two bounding boxes, we use both ② and ③ to obtain accurate positional information and comprehensive image descriptions. For images with more than two annotated bounding boxes, we use ① to achieve comprehensive image descriptions.

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Algorithm 2 A2I

```
Function annotation_to_instruction(dataset, dataset_type):
    prompts ← extract_prompts(dataset, dataset_type)
    instructions ← generate_instructions(prompts, dataset_type)
Function extract_prompts(dataset, dataset_type):
    prompts ← []
    for each image in dataset do
        if dataset_type == 'classification' then
            prompts.append(image['category'])
        else if dataset_type == 'object detection' then
            prompts.append(f"obj[category] [location]")
        else if dataset_type == 'segmentation' then
            bbox_list ← M2B(masks) prompts.append(f"obj[category]
              [location]")
    return prompts
Function generate_instructions(prompts, dataset_type):
    instructions ← []
    for each description in prompts do
        if dataset_type == 'classification' then
            Instruction = "Describe image with [prompt] in detail:"
        else
            if object_number ≤ 2 then
                Instruction1 = "where is/are the [prompt]?"
                Instruction2 = "Describe image with [prompt] in detail:"
            else
                Instruction = "Describe image with [prompt] in detail:"
```

Since Kosmos-2 cannot directly accept bbox or semantic segmentation masks as inputs, we propose the annotation-to-instruction (A2I) algorithm. It automatically converts class labels, object detection bounding boxes, and segmentation masks into instructions that Kosmos-2 can process. The pseudocode is shown in Algorithm 2. These instructions, along with the images, are then fed into Kosmos-2 to generate captions.

We used carefully designed instructions to guide MLLMs in generating accurate and complementary descriptions for RS images. Therefore, we refer to this approach as instruction-guided MLLM generation. Ultimately, we obtained detailed captions with an average of approximately 220 words for each RS image.

Limitations of Stage 1: Although the first stage is designed to maximize visual-semantic coverage by generating descriptions from multiple sources, it still has several limitations. First, the captions produced by different sources are inherently heterogeneous in style, granularity, and reliability. Rule-based descriptions are generally accurate but often lack linguistic naturalness and semantic richness, while MLLM-generated descriptions are more expressive yet may contain redundancy, inconsistencies, or hallucinated content. Second, the generated descriptions are often lengthy and not directly suitable for training models such as CLIP, whose text encoder has a limited token budget. Third, despite the complementary design, the visual perception ability of current MLLMs remains constrained when dealing with complex RS imagery, especially for fine-grained details, small objects, or large-scale scenes. Therefore, the outputs of Stage 1 should be viewed as complementary candidate descriptions rather than final captions. This also motivates the second-stage summarization module, which is introduced to integrate complementary information, reduce noise, and produce more concise and consistent captions. In future work, more advanced vision-language models tailored to high-resolution RS imagery may further improve the quality and reliability of first-stage caption generation.

TABLE III

MANUAL HALLUCINATION ANALYSIS RESULTS. (A) KOSMOS-2 HALLUCINATION DISTRIBUTION. (B) LLAVA 1.6 HALLUCINATION DISTRIBUTION

(a) Kosmos-2 Hallucination Distribution

Category	Ratio (%)
No Hallucination (Accurate)	29.6%
Over-specific Inference	28.0%
Object Presence	14.4%
Count / Scale	10.8%
Spatial Relation /Layout	9.6%
Attributes	4.8%
Others	2.8%

(b) LLaVA 1.6 Hallucination Distribution

Category	Ratio (%)
No Hallucination (Accurate)	35.2%
Over-specific Inference	56.4%
Object Presence	3.0%
Attributes	2.5%
Spatial Relation	2.0%
Others	0.9%

Hallucination Phenomena and Countermeasures: Generative large models inevitably encounter hallucination issues during text generation [42]. To assess the reliability of Kosmos-2 and LLaVA-1.6 in generating captions for RS images, we conducted a systematic human evaluation to investigate hallucination phenomena. We randomly sampled 3000 image-caption pairs (1500 each generated by Kosmos-2 and LLaVA-1.6) from the data generated in the first stage. Five engineers with university degrees were recruited to manually audit these samples, identifying the primary hallucination type for each pair based on a rigorous taxonomy. The evaluation results, detailed in Table III, reveal distinct hallucination profiles for the two models.

For Kosmos-2, only 29.6% of the generated captions were found to be entirely accurate. The model suffers from a fragmented error distribution, most notably over-specific Inference (28.0%) and object presence (14.4%). Significant portions of the data also exhibited count/scale (10.8%) and spatial relation (9.6%) errors, such as describing "rivers running through buildings" where no such layout exists. For LLaVA-1.6, while achieving a higher accuracy rate (35.2%), the model demonstrates a pronounced tendency toward over-specific inference (56.4%). This indicates that while the model is visually sensitive, it frequently "hallucinates" extensive details (e.g., specific but incorrect temporal or locational contexts) that go beyond the actual visual evidence. This is precisely the main reason why the generated text is significantly longer than that generated by other models.

In addition to these dominant patterns, a very small fraction of samples exhibited more erratic behaviors. These include parametric knowledge bias, where the model misinterprets generic scenes as specific world-renowned landmarks, and intrinsic hallucinations, characterized by the generation of garbled text or highly illogical object misidentifications (e.g., mistaking wind turbines for helicopters). Furthermore, occasional misjudgments due to safety mechanisms were observed,

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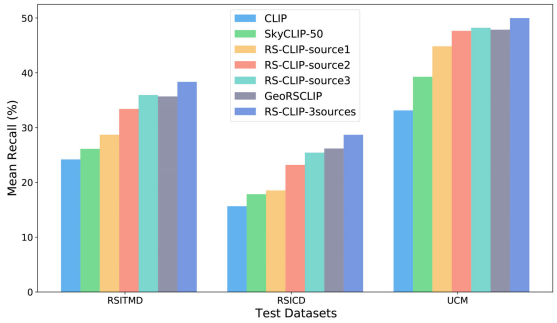


Fig. 4. RS Cross-modal text-image retrieval results of different RS CLIPs fine-tuned with different training data.

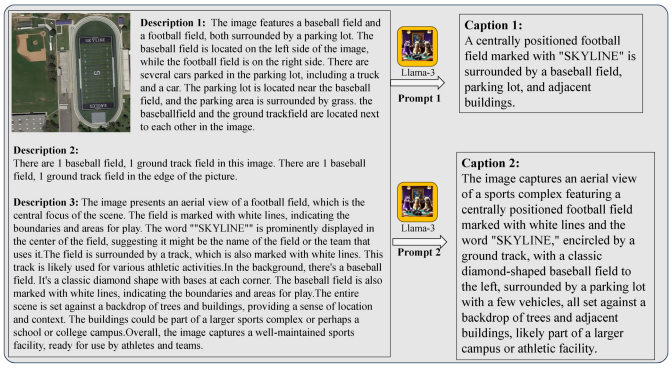


Fig. 5. Pipeline of Stage 2: semantic distillation and alignment-aware compression. Description 1 is generated by Kosmos-2, Description 2 by the R-M Relay method, and Description 3 by LLaVA-1.6. Guided by Prompt 1 and Prompt 2, LLaMA-3 integrates these complementary descriptions and generates compact, aligned captions with different styles. The version of LLaMA-3 used here is LLaMA-3-8B-Instruct.

where the model incorrectly triggered refusals such as "content does not comply with regulations." These rare error types were collectively categorized under the **"Others"** category in our analysis.

To address these issues, we implemented the following strategies. First, to address the issue of misjudgments due to safety mechanisms, we employed regular expressions to filter out garbled text and invalid sentence structures. Then, to resolve the high incidence of overspecific inference and parametric knowledge bias in LLaVA-1.6 (which together account for more than 50% of the errors), we implemented prompt designs in subsequent stages. These prompts guide the LLM to disregard nonvisual information, such as temporal and locational details.

However, it must be acknowledged that these measures provide only partial mitigation of the hallucination issues inherent in generative large models and do not offer a comprehensive solution. Hallucinations are fundamentally tied to the limitations of the models themselves, and current technologies are unable to fully eliminate these issues at their core. Consequently, further research is required to explore additional strategies for addressing this challenge.

Validation of the Effectiveness of Different Descriptions: To validate the effectiveness of our description strategies, we fine-tuned CLIP ViT-B-32 using data from three sources: source1 (R-M Relay), source2 (LLaVA-1.6 generation), and source3 (Kosmos-2 generation). We first trained CLIP on data from each source individually, resulting in RS-CLIP-source1, RS-CLIP-source2, and RS-CLIP-source3. Then, we trained CLIP on data from all three sources combined, resulting in RS-CLIP-3sources. Experimental results in Fig. 4 showed that data from each source significantly improved CLIP's performance on the RS image–text retrieval task, while RS-CLIP-3sources outperformed all previous RS-CLIP models with less training data, demonstrating the complementary nature and effectiveness of these strategies.

C. Stage 2 (Semantic Distillation and Alignment-Aware Compression)

Although the multiperspective descriptions generated in Stage 1 already provide rich and complementary semantic information, they are not yet fully optimized for vision-language alignment. Specifically, rule-based descriptions still lack sufficient linguistic diversity, MLLM-generated descriptions may contain occasional hallucinations or weakly grounded content, and some captions—especially those produced by LLaVA-1.6—are excessively long for models such as CLIP, whose text encoder is constrained by a limited token budget.

To address these limitations, we use LLaMA-3-8B-Instruct to perform semantic distillation over the three descriptions generated in Stage 1 (see Fig. 5). This process consolidates complementary multiperspective information into a unified textual representation for each image while suppressing nonvisual or weakly grounded content and compressing lengthy descriptions into a more compact form. In this way, the resulting captions remain information-rich while becoming better aligned with the input requirements of vision-language models such as CLIP.

When generating distilled captions, LLMs often exhibit a nonunimodal output distribution [43], with multiple peaks or modes. Under a single prompt, the generation process may converge to a local maximum, producing overly averaged descriptions that fail to fully preserve the semantic diversity contained in the multiperspective inputs. To better reflect the complexity of the source descriptions, strategies such as repeated sampling [43] can enhance diversity.

To avoid excessive randomness while still broadening the semantic space, we use different prompts to generate multiple distilled captions with distinct styles and emphases.

Specifically, we generate two captions with distinct styles for each image using two prompts. For one of the two captions, five possible options are generated, from which one is randomly selected for CLIP training. *It is important to clarify that we sampled one out of every five captions for training the CLIP model to explore the effectiveness of the dataset construction strategies. In contrast, the CoCa model was trained using the entire dataset. The final released dataset will also include all available data.*

It should be noted that the multiple captions generated for each image are used as caption candidates rather than being simultaneously fed into CLIP contrastive training. For HQRS-CLIP, we construct a one-to-one image–text training set by

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sampling only one caption for each image. Therefore, different captions corresponding to the same image are not treated as in-batch negatives in the InfoNCE objective during HQRS-CLIP training. In contrast, the full set of captions is retained in the released dataset and is used in CoCa training to improve caption-generation capability.

It is widely recognized that prompts play a crucial role in the performance of LLMs. To maximize the potential of LLMs, we employed the CO-STAR framework [44], which secured the top results in the Prompt Engineering competition organized by the Government Technology Agency of Singapore (GovTech). The **CO-STAR** framework facilitates the construction of effective prompts by considering elements such as **C**ontext, **O**bjective, **S**tyle, **T**one, **A**udience, and **R**esponse. This structured approach enhances the quality and relevance of the responses generated by LLMs. After multiple experiments, we designed two prompts as follows.

Prompt 1: am creating an RS image–text matching dataset. Each image has three different captions provided by different people, describing the same visual content from different perspectives. **Please summarize the main features of the following detailed description in a single sentence.** The sentence should use descriptive language with rich adjectives and prepositional phrases, include specific information about colors and spatial relationships, and maintain a clear and neutral tone. Use a professional and descriptive writing style, with a concise and content-rich tone. The target audience is researchers in the field of RS image analysis. Provide only the extracted sentence without any additional explanations or introductory text.

Prompt 2: I am creating an RS image–text matching dataset. Each image has three different captions provided by different people, describing the same visual content from different perspectives. **Summarize the following three captions into one detailed sentence, including as many key visual elements as possible.** Use a professional and descriptive writing style. The tone should be concise and informative, suitable for researchers in the field of RS image analysis. Generate five detailed descriptions, each focusing on slightly different details to provide complementary information. Each description should be complete and complementary, ensuring full information coverage. **After generating the five detailed descriptions, randomly select one and output only that version.**

Finally, we generated captions with multiple styles for RS images. We fine-tuned CLIP using these data and tested its performance. The experimental results show that both prompts generated high-quality captions, significantly improving the performance of CLIP (see Table XVI).

D. HQRS-It-210K Dataset Description and Human Evaluation

1) Dataset Description: We ultimately constructed the HQRS-IT-210K dataset with 210 556 images. For each image, we used two types of prompts to guide the LLM, generating five captions with one prompt and one caption with the other. This process resulted in a total of six captions per image, bringing the overall number of captions to approximately

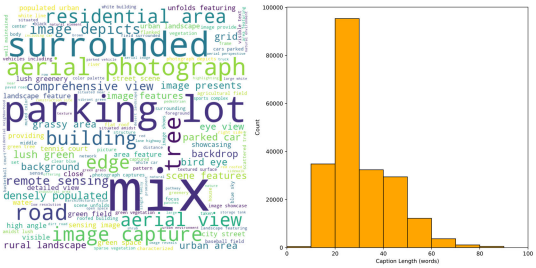


Fig. 6. Word cloud and caption length distribution of the HQRS-210K dataset.

TABLE IV
MANUAL EVALUATION RESULTS ON HQRS-IT-210K. (A) QUALITY SCORES. (B) ERROR TAXONOMY

(a) Quality scores

Score	Percentage
5	27.7%
4	44.6%
3	12%
2	9.4%
1	6.3%

(b) Error taxonomy

Error Type	Percentage
No error	27.7%
Incomplete description	26.7%
Partial mismatch	24.3%
Over-inference	10.0%
visually irrelevant information	9.3%
Redundant description	2.0%

1.26 million. As shown in Fig. 1, the captions are naturally varied in structure and contain rich visual and semantic information. On average, each caption contains approximately 35.6 words, which is significantly longer than the average caption length of 12 words in human-annotated datasets.

As shown in Fig. 6, the word cloud of our captions demonstrates their diversity, descriptive precision, and contextual depth, highlighting the comprehensiveness and professionalism of our captioning approach. The caption length distribution indicates that most captions are rich in information and can be fully utilized by models such as CLIP and CoCa.

2) Manual Quality Evaluation: To systematically assess the image–text matching quality of HQRS-IT-210K, we conducted a manual evaluation on a randomly sampled subset of the dataset. Specifically, 3000 samples were randomly selected from the final dataset, and five engineers with at least a bachelor's degree participated in the evaluation. Each evaluator independently reviewed 600 samples and rated the degree of correspondence between each image and its caption.

A five-point rating scale (1–5) was adopted to measure image–text matching quality, where a score of 5 indicates perfect correspondence. In this case, the caption must accurately and completely describe the visual content of the image without introducing any nonvisual information or omitting key visual elements. In other words, a full score is assigned only when the textual description provides an accurate, complete, and nonredundant representation of the image content. In all other cases, points are deducted according to the severity of the mismatches.

As shown in Table IV(a), even under these strict evaluation criteria, 27.7% of the samples received the full score of 5, indicating a high degree of consistency between the images and their captions. Further inspection reveals that these high-quality samples are typically associated with images containing relatively simple scenes and a limited number of

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visual objects, such as forests, desert landscapes, or images containing only a few ships. Many of these samples originate from classification datasets, where images are often cropped to retain only the primary object category, thereby significantly reducing the complexity of the description task. Since classification datasets account for approximately one-third of the entire dataset, this observation is consistent with the proportion of full-score samples.

In terms of score distribution, 44.6% of the samples received a score of 4, indicating that most samples exhibit good image–text correspondence with only minor omissions or slight deviations in the descriptions. Additionally, 12% of the samples received a score of 3, suggesting moderate inconsistencies between images and captions but still within an acceptable range. Meanwhile, 9.4% of the samples received a score of 2, reflecting noticeable mismatches between image content and textual descriptions. Finally, 6.3% of the samples received the lowest score of 1, indicating severe inconsistencies where the captions fail to accurately reflect the image content. Overall, the average score across all evaluated samples is 3.78, suggesting that HQRS-IT-210K demonstrates generally good image–text alignment and maintains a relatively high level of dataset quality.

3) Manual Error Analysis: Based on the manual quality evaluation, we further analyzed the types of errors present in the dataset. The results show that 27.7% of the samples contain no observable errors and achieve perfect alignment between image content and textual descriptions.

To provide a more fine-grained understanding of the remaining issues, we categorize the observed errors using a taxonomy consisting of incomplete description, partial mismatch, over-inference, visually irrelevant information, and redundant description, as summarized in Table IV(b).

Among the samples containing issues, the most common error type is **incomplete description**, accounting for 26.7% of the samples. In these cases, captions correctly describe certain major elements of the image but fail to cover all important visual information. In RS imagery, smaller objects or targets affected by resolution limitations are particularly prone to being overlooked.

The second most common category is **partial mismatch**, which accounts for 24.3% of the samples. This type of error occurs when certain details in the caption do not match the actual image content, such as incorrect descriptions of object attributes, quantities, or spatial relationships.

In addition, 10.0% of the samples exhibit **over-inference**. In these cases, captions include speculative or inferential statements that go beyond what can be directly observed in the image. For example, a description such as "likely a crop treated with a dye or chemical" introduces assumptions that cannot be verified purely from visual evidence.

Another 9.3% of the samples contain **visually irrelevant information**. These captions include subjective or descriptive expressions that are not directly grounded in the visual content of the image, such as evaluative or stylistic phrases.

Finally, **redundant descriptions** account for 2.0% of the samples. In such cases, captions repeatedly describe the same or similar visual elements, introducing redundancy without adding meaningful information and thereby reducing the overall information density of the text.

Representative examples of these error types are presented in Fig. 7 to provide an intuitive illustration of their manifestations.

Overall, the manual error analysis indicates that the primary issues in HQRS-IT-210K are not widespread severe mismatches but rather incomplete descriptions, localized semantic inaccuracies, and inferential redundancy commonly observed in generated captions. This suggests that although multistage data cleaning and filtering procedures significantly improve dataset quality, large-scale automatically constructed RS image–text datasets are still inevitably influenced by factors such as visual complexity, resolution limitations, and semantic biases in generative models. Building upon the error taxonomy established in this study, our future research will explore targeted quality enhancement strategies to continuously refine the paradigm for the automated construction of large-scale RS image–text datasets.

E. RS Long-Text Image Retrieval Test Set

While testing the retrieval performance of CLIP, we identified a significant limitation in existing RS image–text retrieval test datasets: the captions are too short (approximately 12 words). This leads to two issues: 1) short captions fail to fully describe images, causing category ambiguity and 2) they are inadequate for evaluating CLIP's performance on long-text cross-modal retrieval tasks, as longer captions provide more complex and complete information. To address this limitation, we constructed an RS long-text image retrieval test set.

A simple approach is to concatenate the five short captions into a single long-text caption. However, this approach has major drawbacks: 1) concatenation introduces redundancy and repetition, reducing overall information density. For example, in Fig. 8, Sentences (1) and (3) are semantically similar, while Sentences (4) and (5) are identical. Such repetition dilutes key information and decreases the proportion of effective content and 2) concatenated captions often lack coherence and fluency, as each short caption is originally an independent description, leading to logical disconnections when the captions are linked together.

To address these shortcomings, LLMs can be employed for caption integration and rewriting [43], effectively resolving issues related to redundancy, coherence, and errors (see Fig. 8). First, LLMs enhance information density by eliminating redundancy and repetition through semantic understanding and information extraction, integrating key details from the original captions into a concise and informative long caption. Second, LLMs improve coherence and fluency by connecting independent short sentences with appropriate transitions, resulting in semantically coherent and logically consistent long captions. Third, LLMs correct grammatical and semantic errors in the original captions, which are common because the captions were written by nonnative annotators. For example, in Fig. 8, Sentence (4) and Sentence (5) contain grammatical errors, which are corrected after LLM-based rewriting.

We utilized LLaMA-3-8B-Instruct to combine and rewrite the five captions for each image in the test sets of three

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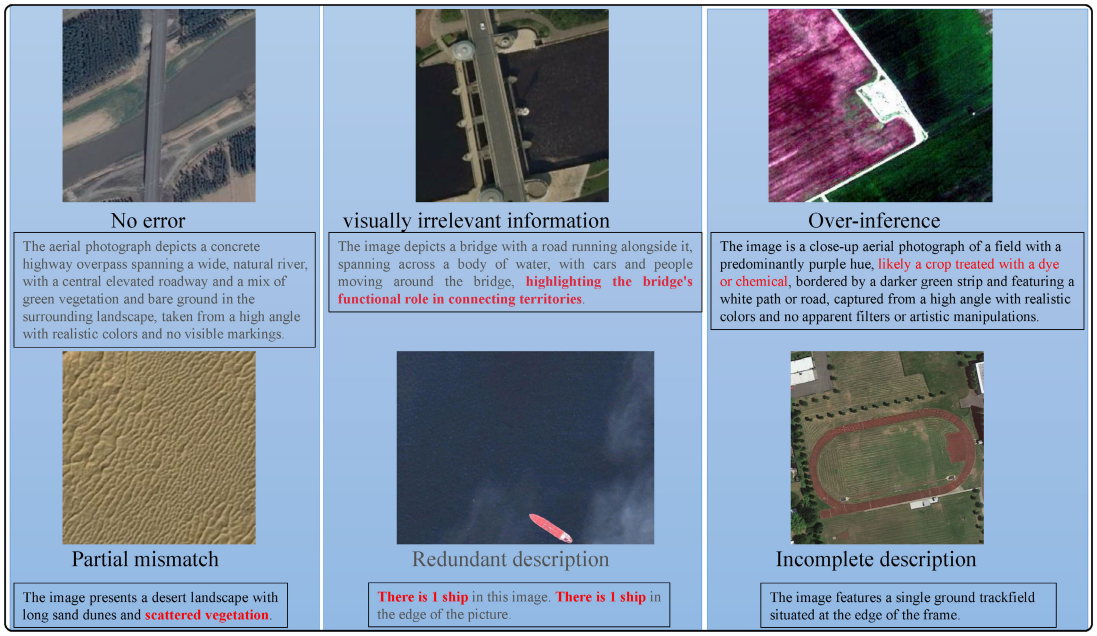


Fig. 7. Examples of different error types in HQRS-IT-210K.

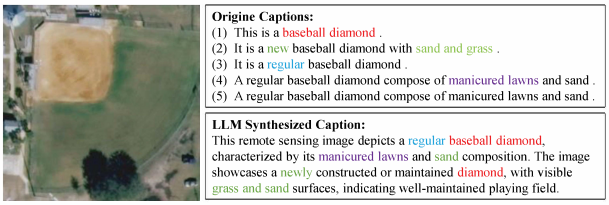


Fig. 8. Example of the original captions in UCM and the corresponding long-caption rewriting result.

widely used benchmark datasets (RSITMD, RSICD, and UCM) into a single long-text caption. This approach has two key advantages: 1) these datasets are well established and widely recognized and 2) most images include five complementary captions, and using an LLM to combine them produces comprehensive, high-quality long-text captions at a low cost.

To ensure that the LLM strictly adheres to the content of the original captions, we carefully designed the following prompt.

Prompt: I am creating an RS image-long text matching dataset based on an existing RS image-short text matching dataset. In the short text matching dataset, each image has 5 different short captions provided by different individuals, describing the same visual content from different perspectives. Please summarize the following 5 short captions into one detailed long caption, ensuring that the information from each short caption is included in the long caption. **The summarization process must strictly adhere to the information given in the 5 short captions without any additional extrapolation.** Use a professional and descriptive writing style, with a concise and content-rich tone. The target audience is researchers in the field of RS image analysis. Do not include any additional explanations or introductory text in your response.

As demonstrated in Fig. 8, the advantages of this approach are evident. The LLM-rewritten long caption effectively integrates key information from each short caption, enhancing the overall information density. This comprehensive description provides a more complete representation of the image, reducing potential category ambiguities. Additionally, by naturally combining sentences, the LLM addresses the coherence and fluency issues that can arise from simple concatenation, ensuring a more seamless and readable caption. Moreover, potential grammatical and semantic errors in the original captions can also be corrected.

For simplicity, we collectively refer to the test sets of the three benchmark datasets as **RET3-test**. The dataset in which the five captions corresponding to each image are concatenated into a "pseudo-long caption" is termed **RET3-test-join**. The dataset with long captions rewritten by the LLM is referred to as **LongRET3-test**. We utilized CLIP ViT-L-14 to calculate the image–text similarity scores for RET3-test, RET3-test-join, and LongRET3-test. These scores were used to evaluate their quality.

Positioning and Limitations of LongRET3-Test: We emphasize that **LongRET3-test** is intended as a supplementary diagnostic benchmark for examining RS image–long-text retrieval under long-caption conditions, rather than as a fully independent benchmark for definitive model comparison. Although LongRET3-test is constructed from the test splits of three widely used RS image–text retrieval datasets, its captions are generated through LLM-assisted rewriting. Since the construction of our training data also involves LLM-assisted long-caption generation and rewriting, the training and evaluation texts may still share certain distributional properties, including similar writing styles, lexical pReferences, sentence structures, and long-form caption organization patterns. Therefore, the results on LongRET3-test should be interpreted as preliminary and diagnostic evidence of long-text

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retrieval behavior rather than conclusive evidence of general superiority across fully independent evaluation settings.

Quality Control for LLM-Rewritten Long Captions: To mitigate the impact of biases and hallucination-related errors introduced during LLM-based rewriting, we adopted both analytical and manual quality-control procedures. First, we systematically analyzed the biases introduced by the rewriting process and their potential impact on retrieval evaluation. Then, we manually reviewed and corrected all rewritten long captions to improve data quality and minimize these adverse effects.

Specifically, we compared the original ground truth (GT) images, the human-annotated short captions, and the LLM-rewritten long captions in the benchmark. Overall, we found that, in most cases, the rewritten long captions successfully integrate visual semantics scattered across multiple short captions, thereby providing richer details, broader semantic coverage, and greater descriptive completeness than the original short annotations. This makes the benchmark more challenging and, in our view, more suitable for evaluating the image–text matching and cross-modal alignment abilities of vision–language models under long-form textual conditions. However, we also observed that LLM-based rewriting may introduce several potential sources of bias that can affect retrieval evaluation.

First, when the original short captions contain overlapping semantics, the rewritten long caption may repeatedly describe the same visual information. While this is usually not a factual error, it increases text length without proportionally increasing effective visual content, thereby reducing semantic density and increasing the burden of long-text encoding and matching. In such cases, degraded retrieval performance may reflect not only weaker vision–language alignment but also limited robustness to redundant long-form input.

Second, during rewriting, the LLM may introduce nonvisual content in the form of modifiers, evaluative expressions, or atmosphere-related language, such as describing a scene as "serene," "peaceful," or "well-maintained." These expressions are not always directly grounded in visible image evidence. As a result, they may occupy semantic space, dilute the density of explicit visual information, and increase the difficulty of processing abstract or implicitly inferred semantics. This means that retrieval performance may be affected by factors beyond pure visual-text alignment.

Third, the LLM may also perform moderate inferential expansion by describing the possible function, role, or contextual meaning of visible objects. For example, a river may be described as potentially serving water supply, transportation, or industrial activity. Although such descriptions may appear linguistically plausible, they go beyond directly observable image content. For models trained mainly on visually grounded alignment rather than on complex long-form descriptions, such added semantics may increase comprehension difficulty and introduce additional challenges for retrieval.

Fourth, in a small number of cases, the original human-written short captions may themselves contain inaccuracies, omissions, or mismatches with the image. Since the LLM in this pipeline mainly performs textual aggregation and rewriting rather than true visual verification, such errors may be inherited and sometimes amplified during rewriting. This may further reduce image–text alignment quality and introduce potential concerns regarding the fairness and reliability of the benchmark.

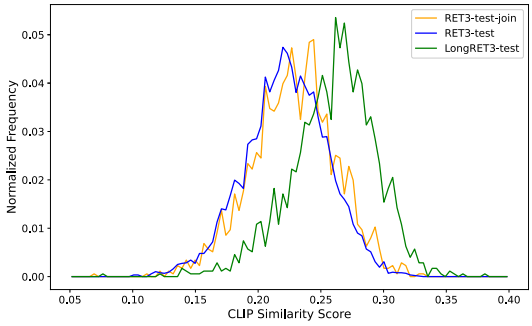


Fig. 9. CLIP similarity score distributions for **RET3-test**, **RET3-test-join**, and **LongRET3-test**.

Based on these observations, LLM-based rewriting introduces both advantages and limitations in benchmark construction. On the one hand, it substantially improves semantic richness, descriptive completeness, and task difficulty by aggregating multiple short captions into a more comprehensive long-form query. On the other hand, it may introduce redundancy, nonvisual content, inferential expressions, and propagated annotation noise.

To mitigate the above biases as much as possible, we introduced a manual quality-control procedure for the rewritten long captions. In particular, the review process focused on four aspects: reducing unnecessary repetition when it did not add new visual information, removing nonvisual or weakly grounded evaluative expressions, correcting excessive inference beyond directly observable image content, and fixing misinformation propagated from flawed original annotations. Each rewritten caption was checked against its corresponding image, and only the parts that were clearly unsupported, overly inferential, or factually inconsistent were revised. Throughout this process, corrections were made conservatively to preserve the overall semantic richness and long-form structure of the captions while improving their visual grounding and reliability.

After this manual quality-control process, we further examined the impact of our revisions on the image–text similarity distribution. As illustrated in Fig. 9, the distribution of image–text similarity scores for the original RET3-test closely mirrors that of RET3-test-join. LongRET3-test not only exhibits a distribution similar to those of the first two datasets but also demonstrates an overall increase in similarity scores. The primary reason for this improvement is that combining the information from the five distinct captions reduces category ambiguity, resulting in captions that are more accurately aligned with the image content.

IV. Experiments

To evaluate the effectiveness of our dataset on both discriminative and generative tasks, we fine-tuned the CLIP and CoCa models separately. For the CLIP model, we used only 1/6 of the entire dataset, with one caption per image, to highlight the

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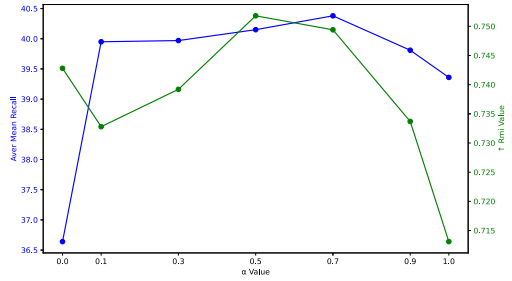


Fig. 10. Effectiveness of different α values.

high quality of our dataset while maintaining comparability with similar studies. In contrast, for the CoCa model, due to the higher data demands of generative tasks, we utilized the full dataset of approximately 1.26 million captions for training. Since similar studies primarily focus on the CLIP model, this article also emphasizes the study and comparison of the CLIP model.

A. Style-Level Caption Sampling Strategy for Clip Training Data

When constructing the training data for CLIP, we adopted a simple yet effective style-level caption sampling strategy. In the second stage, we used two prompts to guide the LLM to generate two different caption styles, aiming to simulate the diversity of real-world descriptions. For Prompt 1, the LLM generated five captions, from which one was randomly selected. Each image was associated with two captions of different styles generated from two distinct prompts.

Instead of directly concatenating or semantically merging these two captions into one longer caption, we assigned each image only one training caption by randomly selecting one of the two candidates according to the probability α. This one-caption-per-image sampling strategy keeps the CLIP training set in a one-to-one image–text format and avoids false negatives caused by multiple captions of the same image appearing in the same contrastive batch. α denotes the proportion of captions generated by Prompt 2 in the final training set, and $\alpha = 0$ means that only Prompt 1 captions were used.

Experimental results show that this simple style-level sampling strategy significantly improves model performance without increasing training costs and outperforms models trained on any single caption style (see Fig. 10).sity are related but not equivalent factors.

The effectiveness of this strategy can be understood from three aspects. First, the two prompts encourage the LLM to focus on different aspects of the same image, producing complementary captions with different emphases, styles, and semantic coverage. Second, mixing caption styles increases dataset-level diversity, which helps the model generalize better and reduces overfitting to a single caption pattern. Third, this strategy improves semantic coverage at the dataset level without increasing the length of each individual training caption. Compared with directly concatenating multiple captions, it avoids excessive redundancy, ambiguity, and token-budget pressure, thereby preserving a better balance between semantic diversity, compactness, and effective information density.

B. Implementation Details of Experiments

CLIP training: In RemoteCLIP, SkyCLIP, and GeoRSCLIP, the models were trained using the full fine-tuning approach. To ensure a fair comparison, we also used this method. The CLIP ViT-B-32 and CLIP ViT-L-14 models were fully fine-tuned on our dataset. To emphasize the intrinsic effectiveness of our dataset, we did not use any data augmentation techniques or perform any specific hyperparameter tuning during training. We randomly selected 15% of the data as the validation set, with the remaining data used for training. The training process utilized a cosine learning rate scheduler, mixed-precision (AMP) mode, and the AdamW optimizer [45]. Cross-modal interaction was optimized using the InfoNCE loss [46]. For CLIP training, we used the sampled one-to-one image–text subset described above, where each image is paired with only one caption. Therefore, the standard InfoNCE loss is applied to one-to-one image–text pairs, and captions of the same image do not appear as negatives to each other. For ViT-B-32, the learning rate was set to $2e^{-5}$, and the batch size was set to 256. For ViT-L-14, the learning rate was set to $1e^{-6}$, and the batch size was set to 32. Ultimately, we obtained HQRS-CLIP (ViT-B-32) and HQRS-CLIP (ViT-L-14). **Compared with RemoteCLIP's 233.4 h of training time, training our HQRS-CLIP (ViT-L-14) required only approximately 1.5 h.**

CoCa Training: The CoCa model was also trained using full fine-tuning. We utilized all 1.26 million image–text pairs without applying any data augmentation strategies. The AdamW optimizer was also used during training, with the CoCa-contrastive-loss weight set to 1 and the CoCa-caption-loss weight set to 2. We trained only the CoCa ViT-L-14 model with a learning rate of $2e^{-5}$ and a batch size of 32. During benchmark testing, we utilized the training data from the corresponding benchmark datasets and adjusted the loss weights by setting the CoCa-contrastive-loss weight to 0 and the CoCa-caption-loss weight to 1. This adjustment was made to enhance the model's caption generation capability for the respective datasets.

For CoCa training, we followed the commonly adopted pair-level random sampling protocol [3], [5] for large-scale image–text pretraining, which is also consistent with the prior RS vision-language studies used for comparison. Under this setting, multiple captions of the same image are treated as separate image–text pairs in the dataset. Therefore, in principle, another caption of the same image may appear in the same mini-batch and be regarded as an in-batch negative by the contrastive branch. However, this event is statistically rare in our setting. HQRS-IT-210K contains 210 556 images and six captions per image, resulting in 1 263 336 image–text pairs. With a batch size of 32, the probability that a given anchor encounters at least one caption from the same image among its in-batch negatives is approximately

$$P_{\mathrm{FN}} = 1 - \frac{\binom{1,263,330}{31}}{\binom{1,263,335}{31}} \approx 0.0123\%. \quad (1)$$

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Equivalently, the expected proportion of false-negative off-diagonal entries in the contrastive similarity matrix is only

$$R_{\mathrm{FN}} = \frac{5}{1,263,335} \approx 0.000396\%. \tag{2}$$

Thus, such false negatives are extremely sparse and are unlikely to substantially affect the overall contrastive optimization.

Both the CLIP and CoCa models were trained on a single RTX 4090 24 GB GPU, and the training code for both models was based on OpenCLIP.[1]

We conducted model testing on the following tasks.

HQRS-CLIP [RS Cross-modal Text–Image Retrieval (RSCTIR)]: This task encompasses both text-to-image and image-to-text retrieval using the UCM, RSICD, and RSITMD datasets. We also assessed zero-shot retrieval to test generalization to unseen datasets. Performance was measured using R@1 for both retrieval directions and mean recall.

HQRS-CLIP (RS Cross-modal Long-Text Image Retrieval): To better evaluate the model's ability to understand longer and more comprehensive descriptions, we created the first RS long-text image retrieval test set by merging and rewriting existing datasets. The new captions average 72 words, significantly longer than the typical 12-word captions, and offer improved alignment with images. We evaluated performance using i2t-R@1, i2t-R@5, i2t-R@10, t2i-R@1, t2i-R@5, t2i-R@10, and mean recall.

HQRS-CLIP [SeLo]: SeLo is defined as the use of cross-modal information to locate semantically similar regions in large-scale RS scenes and is considered a more advanced retrieval task than RSCTIR. AIR-SLT is currently the only large-scale multimodal SeLo test set. We employed four metrics: Rsu (Significant Area Proportion), Ras (Attention Shift Distance), Rda (Discrete Attention Distance), and Rmi (a composite metric), to comprehensively assess performance.

HQRS-CLIP (ZSC): Following the extended definition of zero-shot learning, we tested the model's ability to generalize to unseen datasets using AID, RESISC45, and EuroSAT, with top-1 accuracy as the evaluation metric. These three datasets were also tested by models such as SkyCLIP, GeoRSCLIP, and RemoteCLIP. We selected them for testing to ensure a fair comparison with these models.

HQRS-CLIP (FSC): To assess the model's capability with limited labeled data, we conducted 5-way 1-shot and 5-way 5-shot classification on the RESISC45 and UCM datasets, based on the benchmark introduced in [47]. This evaluation assessed the feature extraction performance of the RS CLIP image encoder under few-shot conditions.

RS-CoCa (RSIC): For the RS-CoCa model, we evaluated its performance on the RSIC task. We used the widely adopted RSICD and UCM-caption datasets as benchmarks to facilitate comparison with other studies. Similarly, we employed commonly used evaluation metrics, including BLEU, METEOR, ROUGE-L, CIDEr, and SPICE. Additionally, to provide a more intuitive demonstration of our model's strong performance, we conducted a qualitative analysis by presenting examples of captions generated by RS-CoCa.

[1]https://github.com/mlfoundations/open_clip

TABLE V

RESULTS OF ZERO-SHOT RSCTIR TASK. THE BEST RESULT IS IN BOLD. **TRAINING PAIRS** INDICATES THE NUMBER OF TRAINING IMAGE–TEXT PAIRS USED FOR TRAINING THE RS CLIP MODELS

Test Dataset	Models	Training Pairs	I2T R@1	T2I R@1	Mean Recall
RSITMD	CLIP	-	9.51	8.81	24.19
	SkyCLIP-50	1.3 Million+	8.63	11.02	26.12
	GeoRSCLIP	5 Million+	19.03	14.16	36.68
	Ours	**210K+**	**20.58**	**17.30**	**40.15**
	Ours (ViT-L-14)	**210K+**	**21.90**	**19.16**	**40.63**
RSICD	CLIP	-	5.31	5.78	15.74
	SkyCLIP-50	1.3 Million+	5.86	6.31	17.85
	GeoRSCLIP	5 Million+	11.53	9.52	26.18
	Ours	**210K+**	**15.55**	**11.20**	**29.37**
	Ours (ViT-L-14)	**210K+**	**15.55**	**11.95**	**30.33**
UCM	CLIP	-	9.52	8.67	33.13
	SkyCLIP-50	1.3 Million+	10.00	9.90	39.27
	GeoRSCLIP	5 Million+	18.57	13.81	47.76
	Ours	**210K+**	**22.86**	**16.19**	**51.59**
	Ours (ViT-L-14)	**210K+**	22.38	**16.67**	**53.33**

C. Experimental Results

For RS-CoCa, we primarily evaluated its performance on the RSIC task to facilitate comparisons with similar VLFMs. The image backbone of RS-CoCa is ViT-L-14. For HQRS-CLIP, we evaluated its performance on a series of vision-language tasks. The image backbone of HQRS-CLIP is ViT-B-32 unless otherwise specified. All the test sets have been used in previous related RS CLIP studies, and we use the same test sets to ensure a fair comparison. *Experimental results indicate that HQRS-CLIP shows superior capabilities while using only 4.2% of the data used by GeoRSCLIP, 16% of the data used by SkyCLIP-50, and 25% of the data used by RemoteCLIP.*

1) RS Cross-Modal Text-Image Retrieval: **Zero-shot RSCTIR task**. Table V shows that HQRS-CLIP surpasses existing RS CLIP models on 3 benchmarks in the zero-shot RSCTIR setting. Specifically, HQRS-CLIP achieves an average mean recall that is 16.02% higher than that of the CLIP baseline, 12.62% higher than that of SkyCLIP-50, and 3.50% higher than that of GeoRSCLIP. Additionally, scaling up to CLIP ViT-L-14 further improves performance, increasing the average mean recall by 1.06% compared with HQRS-CLIP ViT-B-32, demonstrating the benefits of larger models in leveraging our dataset.

Fine-Tuned RSCTIR Task: For a fair comparison with RemoteCLIP and GeoRSCLIP, we fine-tuned HQRS-CLIP on RET-3.[2] As shown in Table VI, HQRS-CLIP achieved the best results across the majority of metrics. Notably, HQRS-CLIP excels in the Text-to-Image retrieval task. Overall, HQRS-CLIP outperforms RemoteCLIP by 2.94% and GeoRSCLIP by 2.26% in average mean recall. Furthermore, HQRS-CLIP (ViT-L-14) exceeds RemoteCLIP (ViT-L-14) by 4.67%, establishing it as the state-of-the-art model on all three benchmarks.

2) RS Cross-Modal Long-Text Image Retrieval: Before reporting these experimental results, it is important to note that LongRET3-test is used as a supplementary diagnostic benchmark rather than a fully independent benchmark for

[2]RET-3 provided by RemoteCLIP includes 68 565 deduplicated image–textpairs from RSITMD, RSICD, and UCM.

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TABLE VI

RESULTS OF FINE-TUNED RSCTIR TASKS. THE BEST RESULT IS IN BOLD. **TRAINING PAIRS** INDICATES THE NUMBER OF TRAINING IMAGE–TEXT PAIRS USED FOR TRAINING THE RS CLIP MODELS

Image Backbone	Test Dataset	Models	Training Pairs	I2T R@1	T2I R@1	Mean Recall
ViT-B-32	RSITMD	RemoteCLIP	820 K	27.88	22.17	49.38
		GeoRSCLIP	5 Million+	30.09	23.54	50.10
		Ours	**210K+**	**35.18**	**29.65**	**54.53**
	RSICD	RemoteCLIP	820 K+	17.02	13.71	35.26
		GeoRSCLIP	5 Million+	**22.14**	15.26	38.00
		Ours	**210K+**	20.86	**15.61**	**38.08**
	UCM	RemoteCLIP	820 K+	20.48	18.67	56.36
		Ours	**210K+**	**22.86**	**19.14**	**57.21**
ViT-L-14	RSITMD	RemoteCLIP	820 K+	28.76	23.76	50.52
		Ours	**210K+**	**34.07**	**31.02**	**57.05**
	RSICD	RemoteCLIP	820 K	18.39	14.73	36.35
		Ours	**210K+**	21.5	**18.54**	**40.19**
	UCM	RemoteCLIP	820 K	19.05	17.71	54.68
		Ours	**210K+**	20.48	**19.52**	**58.32**

TABLE VII

RESULTS OF LONGRET3-TEST RETRIEVAL. THE BEST RESULT IS IN BOLD. **FT** MEANS THAT THE MODEL WAS FINE-TUNED ON THE RET-3 TRAINING DATA

Model	I2T-R@1	I2T-R@5	I2T-R@10	T2I-R@1	T2I-R@5	T2I-R@10	Mean-Recall
CLIP-Baseline	4.79	16.01	24.44	3.25	14.13	22.91	14.25
SkyCLIP-50	5.01	16.13	27.12	5.41	16.92	28.21	16.47
GeoRSCLIP	11.40	30.48	43.82	9.86	27.12	38.92	26.93
Ours	**14.70**	**34.93**	**48.43**	**15.84**	**37.66**	**51.68**	**33.87**
RemoteCLIP(FT)	12.71	35.44	52.02	9.86	30.54	45.41	31.00
Ours(FT)	**16.58**	**44.33**	**60.57**	**16.01**	**40.85**	**55.84**	**39.03**

definitive comparison. Since both the proposed training data construction and LongRET3-test involve LLM-assisted long-caption generation or rewriting, there may exist distributional similarities between training and evaluation texts in terms of writing style, lexical patterns, and long-caption organization. Therefore, the following results should be interpreted as preliminary evidence for analyzing RS image–long-text retrieval behavior rather than as fully conclusive evidence of overall model superiority.

We evaluated the existing RS CLIP models on the LongRET3-test dataset, with the results presented in Table VII. Among the models not fine-tuned on RET-3, HQRS-CLIP outperformed SkyCLIP-50 by 17.4% and GeoRSCLIP by 6.94% in mean recall. For models fine-tuned on RET-3, HQRS-CLIP achieved a mean recall that is 8.03% higher than that of RemoteCLIP. Moreover, whether fine-tuned on RET-3 or not, HQRS-CLIP consistently achieved the best recall scores across all metrics under this diagnostic setting. These results suggest that HQRS-CLIP has a stronger capability to handle RS image–long-text retrieval cases in LongRET3-test. However, considering the LLM-assisted construction of both training and evaluation texts, these results should be regarded as complementary evidence rather than definitive proof of general superiority on fully independent long-text retrieval benchmarks.

3) SeLo Task: Table VIII demonstrates that HQRS-CLIP outperforms all existing models on the AIR-SLT dataset across all four metrics. HQRS-CLIP achieves the highest Rsu, indicating maximum attention to the GT region, the lowest Ras, reflecting minimal attention shift distance, and the best Rda, showing highly concentrated attention within the GT region. Additionally, HQRS-CLIP leads in the Rmi score, highlighting its exceptional performance in SeLo.

TABLE VIII

RESULTS OF SELO TASK. THE BEST RESULT IS IN BOLD

Method	Training pairs	Rsu ↑	Ras ↓	Rda ↑	Rmi ↑
CLIP-Baseline	-	0.7188	0.3006	0.6992	0.7071
SkyCLIP-50	1.3 Million+	0.7341	0.2997	0.6948	0.7125
RemoteCLIP	820K+	0.7365	0.3008	0.6928	0.7125
GeoRSCLIP	5 Million+	0.7546	0.2610	0.7180	0.7400
ours	**210K+**	**0.7635**	**0.2488**	**0.734**	**0.7518**

TABLE IX

RESULTS OF ZSC TASKS. THE BEST RESULT IS IN BOLD

Method	Training Pairs	AID	RESISC45	EuroSAT	Ave
CLIP-Baseline	-	66.22	60.90	47.21	58.11
SkyCLIP-50	1.3 million	69.49	62.92	49.62	60.68
RemoteCLIP	820K	**92.87**	70.84	35.01	66.24
GeoRSCLIP	5 Million+	73.72	71.89	**61.49**	69.03
Ours	210 K	73.86	**78.41**	60.55	**70.94**

4) Zero-Shot Classification: Since our full dataset includes portions of these three datasets, we excluded overlapping data from our full dataset before independently training CLIP for evaluation to prevent potential data leakage and ensure an unbiased comparison.

As shown in Table IX, HQRS-CLIP achieved the highest average zero-shot top-1 accuracy across the three datasets, outperforming GeoRSCLIP by 1.91%, RemoteCLIP by 4.70%, and SkyCLIP-50 by 10.26%. Additionally, RemoteCLIP displayed inconsistent performance, achieving 92.87% on the AID dataset but performing poorly on EuroSAT, likely due to training data imbalance. In contrast, HQRS-CLIP and GeoRSCLIP demonstrated similar robustness across all three datasets.

5) Few-Shot Classification: Following the benchmark in [47], we conducted two types of FSC tests, as shown in Table X. The first type involved testing on a data subset, the entire dataset, and all available classes. The second type comprised four settings: Setting A tested only the test subset without using the training data; Setting B evaluated all classes within the test subset, consisting of 10 and 6 classes for the two datasets, respectively; Setting C used the entire dataset with five randomly selected classes for each task; and Setting D tested the entire dataset with all classes distinguished, featuring 45 and 21 classes for the two datasets, respectively. Overall, Setting A aligns with standard practice, while Setting D represents a more extreme and challenging scenario involving larger datasets and more classes.

Since our full dataset includes portions of the RESISC45 data, we first excluded all RESISC45 entries before fine-tuning CLIP to prevent data leakage and ensure an unbiased comparison. The experimental results demonstrate that HQRS-CLIP outperformed previous RS CLIP models in nearly all settings (15/16). Notably, in the 1-shot tasks, HQRS-CLIP significantly surpassed all existing RS CLIP models. These findings indicate that our image encoder has developed strong feature recognition and generalization capabilities on our dataset, indirectly validating the high quality of our dataset.

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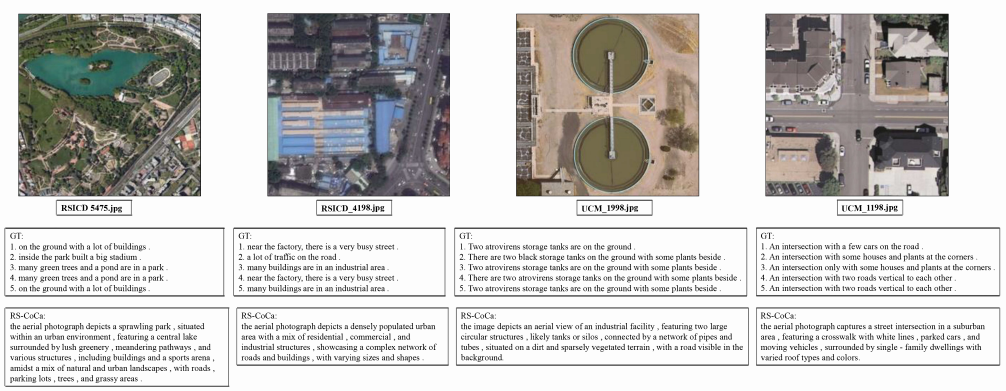


Fig. 11. Sample captions generated by RS-CoCa.

TABLE X
RESULTS OFN-WAY K-SHOT CLASSIFICATION ACCURACY. THE BEST RESULT IS IN BOLD

test set	method	way	RESISC45		way	UCM	
			1-shot	5-shot		1-shot	5-shot
test split	*baseline*	5	86.55 ± 0.18	91.06 ± 0.11	5	70.73 ± 0.16	82.92 ± 0.12
test split (Setting A)	RemoteCLIP	5	87.11±0.08	95.18±0.03	5	-	-
	GeoRSCLIP	5	91.57±0.05	**97.08±0.02**	5	91.95±0.06	98.69±0.01
	SkyCLIP	5	84.37±0.08	94.10±0.03	5	93.39±0.04	98.83±0.01
	Ours	5	**92.15±0.05**	96.58±0.03	5	**96.15±0.03**	**99.54±0.00**
test split (all classes) (Setting B)	RemoteCLIP	10	81.44±0.04	91.91±0.01	6	-	-
	GeoRSCLIP	10	82.32±0.05	93.25±0.01	6	92.23±0.04	98.29±0.01
	SkyCLIP	10	75.45±0.05	89.82±0.01	6	90.99±0.05	98.41±0.01
	Ours	10	**85.09±0.03**	**94.14±0.01**	6	**95.48±0.02**	**99.38±0.00**
whole dataset (all classes) (Setting C)	RemoteCLIP	5	87.01±0.09	96.01±0.04	5	-	-
	GeoRSCLIP	5	91.10±0.06	97.35±0.02	5	92.27±0.06	96.43±0.04
	SkyCLIP	5	86.05±0.08	95.74±0.03	5	91.67±0.06	97.01±0.03
	Ours	5	**93.05±0.05**	**97.82±0.02**	5	**95.23±0.05**	**97.64±0.04**
whole dataset (all classes) (Setting D)	RemoteCLIP	45	62.43±0.03	81.18±0.01	21	-	-
	GeoRSCLIP	45	66.24±0.03	84.24±0.01	21	77.05±0.03	91.04±0.01
	SkyCLIP	45	57.12±0.03	78.42±0.01	21	72.61±0.04	89.26±0.01
	Ours	45	**68.71±0.03**	**85.83±0.01**	21	**82.42±0.02**	**93.06±0.01**

6) RS Image Captioning: Qualitative Analysis: As shown in Fig. 11, the captions generated by RS-CoCa exhibit exceptionally high quality and provide detailed semantic descriptions of RS images. Compared with human annotations, RS-CoCa captions are substantially longer and often contain richer contextual information. This trend is also reflected in the CLIP similarity score distributions (Fig. 12), where RS-CoCa shows higher image-caption similarity. Additionally, the average length of RS-CoCa captions is 48.88 words, significantly exceeding the 10.34 words of human annotations, thereby providing much richer information. These comprehensive and precise descriptions not only enhance the interpretation of RS images but also facilitate more effective image–text alignment during the training of VLMs.

Quantitative Analysis: Table XI illustrates the performance of various models on the RSICD and UCM datasets across multiple metrics. Notably, RS-CoCa achieves state-of-the-art results or results comparable to those of leading methods in semantic relevance and sentence diversity metrics, including METEOR, ROUGE_L, CIDEr, and SPICE. This underscores RS-CoCa's capability to effectively generate rich and diverse descriptive captions. In contrast, SkyEyeGPT exhibits superior performance in BLEU metrics, which can be attributed to its extensive use of manually annotated training data. This reliance results in a high degree of similarity between the training and test data distributions, thereby enhancing its performance on BLEU scores. Despite this, RS-CoCa remains strongly competitive in BLEU metrics, demonstrating its robust overall performance.

Human Evaluation: To systematically assess the quality of captions generated by RS-CoCa, we recruited three engineers with at least undergraduate-level educational backgrounds to conduct a manual evaluation on 1094 samples from the RSICD test set. The evaluation focused on two dimensions: *accuracy* and *completeness*, comparing RS-CoCa-generated captions against GT captions.

Specifically, for each test image, we presented the evaluators with one caption generated by RS-CoCa and one caption randomly sampled from the corresponding GT annotations. The evaluators were asked to independently determine, based on the image content, which caption more accurately described the visual elements in the image (*accuracy*) and which caption more comprehensively covered the visual information in the image (*completeness*). All three evaluators independently compared the two captions for every sample and provided a binary judgment for each dimension.

To further assess the reliability and statistical significance of the pairwise preference evaluation, we conducted additional statistical analysis. Since each test image corresponds to a paired comparison between an RS-CoCa-generated caption

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TABLE XI
RESULTS OF RS CAPTIONING TASK. THE BEST RESULT IS IN BOLD

Dataset	Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr	SPICE
RSICD	BITA	0.774	0.665	0.577	0.504	0.420	**0.717**	**3.045**	0.548
	RSGPT	0.703	0.542	0.440	0.368	0.301	0.533	1.029	NA
	SkyEyeGPT	**0.867**	**0.767**	**0.673**	**0.600**	0.354	0.626	0.837	NA
	RS-CapRet	0.741	0.622	0.529	0.455	0.376	0.649	2.605	0.484
	Ours	0.757	0.626	0.527	0.449	**0.425**	0.708	2.517	**0.555**
UCM	BITA	0.889	0.831	0.773	0.719	0.469	0.838	**3.845**	0.549
	RSGPT	0.861	0.791	0.723	0.657	0.422	0.783	3.332	NA
	SkyEyeGPT	**0.907**	**0.857**	**0.816**	**0.784**	0.462	0.795	2.368	NA
	RS-CapRet	0.833	0.760	0.699	0.645	0.447	0.786	3.429	0.525
	Ours	0.890	0.834	0.783	0.734	**0.487**	**0.842**	3.701	0.539

TABLE XII
HUMAN EVALUATION RESULTS COMPARING RS-COCA CAPTIONS WITH GROUND-TRUTH CAPTIONS ON THE RSICD TEST SET. THE VALUES INDICATE THE PERCENTAGE OF SAMPLES FOR WHICH RS-COCA WAS JUDGED SUPERIOR

Dimension	Evaluator 1 (%)	Evaluator 2 (%)	Evaluator 3 (%)
Accuracy	90.86	89.21	79.98
Completeness	97.99	95.16	92.60

TABLE XIII
PAIRWISE PREFERENCE EVALUATION BETWEEN RS-COCA CAPTIONS AND RSICD CAPTIONS ON THE RSICD TEST SET. PREFERENCE RATES ARE COMPUTED BY MAJORITY VOTING OVER THREE INDEPENDENT EVALUATORS. STATISTICAL SIGNIFICANCE IS ASSESSED USING ONE-SIDED EXACT BINOMIAL SIGN TESTS UNDER $H_0 : p = 0.5$. THE 95% CONFIDENCE INTERVALS ARE ESTIMATED USING NONPARAMETRIC BOOTSTRAP RESAMPLING OVER IMAGES WITH 10 000 ITERATIONS

Dimension	RS-CoCa Preferred	Preference Rate	95% Bootstrap CI	p-value	Fleiss' κ
Accuracy	982 / 1094	89.76%	87.93–91.50%	1.75×10^{-174}	0.327
Completeness	1058 / 1094	96.71%	95.61–97.71%	1.86×10^{-262}	0.314

and an RSICD GT caption, we performed one-sided exact binomial sign tests under the null hypothesis that RS-CoCa and RSICD GT captions are equally likely to be preferred, i.e., $H_0 : p = 0.5$. To avoid treating multiple judgments for the same image as independent samples, we aggregated the three evaluator judgments for each image using majority voting and treated each image as one paired comparison unit. We also estimated 95% confidence intervals using nonparametric bootstrap resampling over images with 10 000 iterations.

As shown in Table XII, all three evaluators considered RS-CoCa captions to be more accurate than GT captions in the majority of cases. For the accuracy dimension, the proportions of samples on which RS-CoCa was judged superior were 90.86%, 89.21%, and 79.98%, respectively. The advantage of RS-CoCa was even more pronounced in terms of completeness, with corresponding proportions of 97.99%, 95.16%, and 92.60%. These findings indicate that RS-CoCa not only captures the core visual content of RS images with high accuracy but also provides substantially more comprehensive and semantically richer descriptions than the corresponding RSICD GT annotations in this pairwise evaluation setting.

Table XIII further reports the majority-vote preference results, confidence intervals, statistical significance tests, and interrater agreement. Using majority voting across the three human evaluators, RS-CoCa captions were preferred over RSICD GT captions in 982 out of 1094 samples for accuracy, corresponding to a preference rate of 89.76% with a 95% bootstrap confidence interval of 87.93%–91.50%. The one-sided exact binomial sign test shows that this preference is statistically significant ($p = 1.75 \times 10^{-174}$). For completeness, RS-CoCa captions were preferred in 1058 out of 1094 samples, corresponding to a preference rate of 96.71% with a 95% bootstrap confidence interval of 95.61%–97.71%, and the preference is also statistically significant ($p = 1.86 \times 10^{-262}$).

Regarding interrater agreement, the full agreement rates are 76.69% for accuracy and 90.68% for completeness. The corresponding Fleiss' κ [48] values are 0.327 and 0.314, respectively, indicating fair agreement after correcting for chance. We therefore interpret the interrater agreement conservatively as fair rather than strong. Although the chance-corrected agreement is not strong, the majority-vote preference rates, bootstrap confidence intervals, and exact binomial sign tests consistently demonstrate a statistically significant preference for RS-CoCa captions over RSICD GT captions in both evaluation dimensions.

D. Ablation Study of Dataset Construction Method

We chose CLIP for the dataset construction ablation studies primarily because of its simplicity, computational efficiency, and focus on contrastive learning, which allow for more interpretable evaluations of the impact of different factors on multimodal alignment performance. Compared with the more complex CoCa model, which emphasizes generative tasks, CLIP is better suited for such experiments. We conducted ablation studies using cross-modal retrieval and SeLo tasks, as they are multimodal tasks that more effectively evaluate CLIP's multimodal alignment capabilities and reveal the impact of multimodal data and strategies.

To intuitively reflect the overall retrieval performance of the CLIP model, we use **3ret** to represent the RSITMD, RSICD, and UCM test sets. The performance of the RS-CLIP model on 3ret is averaged, with Aver I2T R@1, Aver T2I R@1, and Aver Mean Recall representing the average values of the corresponding metrics across the three test sets.

1) Effectiveness of Different Description Methods: Captions that describe only a portion of an image's visual elements can cause category ambiguity [16], [17], leading to unclear positive and negative pairs during training and reducing the model's recognition capabilities. To test this, we trained two models using the same images with different captions: one

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TABLE XIV

COMPARISON OF GENERATION METHODS ON RETRIEVAL AND SELO TASKS. THE BEST RESULT IS IN BOLD. COMP MEANS COMPREHENSIVE. AVG MEANS AVERAGE

Method	Test	Avg Recall @1		Avg	Semantic Localization			
		I2T	T2I	Mean Recall	↑ Rsu	↓ Ras	↑ Rda	↑ Rmi
Partial	3ret	12.64	10.47	32.49	0.7346	0.3125	**0.6953**	0.7083
Comp	3ret	**17.78**	**13.18**	**37.40**	**0.7526**	**0.2805**	0.6909	**0.7256**

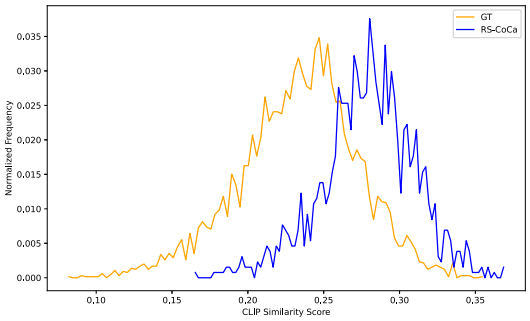


Fig. 12. CLIP similarity score distributions for RS-CoCa captions and GT captions on the benchmark dataset.

with partial descriptions extracted by an LLM (RS-CLIP-part) and the other with comprehensive descriptions (RS-CLIP-full).

As shown in Table XIV, the model trained with fully descriptive captions significantly outperformed the model trained with partial descriptions in retrieval performance. In the SeLo task, the Rsu and Rmi metrics were notably higher, while Ras was lower, for RS-CLIP-full. These results indicate that providing complete captions for each image is essential for enhancing model performance.

2) Effectiveness of Different Data Scales: To investigate the impact of different numbers of training images on performance, we sampled 1/16, 1/8, 1/4, 1/2, and all of the total images to train five models, named RS-CLIP-1 through RS-CLIP-5. The test results are shown in Fig. 13.

As shown in Fig. 13(a), retrieval performance consistently improves with larger image datasets, indicating that more training data enhances retrieval capabilities. However, Fig. 13(b) reveals that the SeLo task metrics do not steadily increase with dataset size. The metrics increase up to the 1/4 scale (approximately 50K images) but plateau or decline beyond that scale. Additionally, Rsu values remain consistently high across all scales. These results suggest that increasing the number of images does not necessarily improve SeLo performance.

3) Effectiveness of Different Caption Lengths: The study in [15] demonstrated that condensing lengthy captions into concise descriptions of key visual information can significantly enhance CLIP's performance. However, the effect of varying caption lengths on CLIP has not been thoroughly explored. In our experiment, we used LLMs to create captions of different lengths from the same descriptive information and trained CLIP models accordingly.

Fig. 14(a) shows that retrieval performance improves as the average caption length increases to approximately 50 words, and Fig. 14(b) reveals a similar trend for the SeLo task. Beyond this range, longer captions lead to a decline in performance because many captions exceed CLIP's 77-token limit, causing text truncation and information loss. These results suggest that performance does not increase monotonically with caption length. Therefore, the gain does not come from longer text alone; rather, it depends on whether the added text introduces useful complementary semantics or merely dilutes the effective visual information density. In this sense, caption length and information density are related but not equivalent factors.

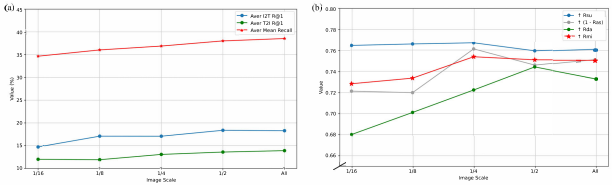


Fig. 13. Effectiveness of different data scales. (a) Image retrieval performance under different training dataset scales. (b) Semantic localization performance under different training dataset scales.

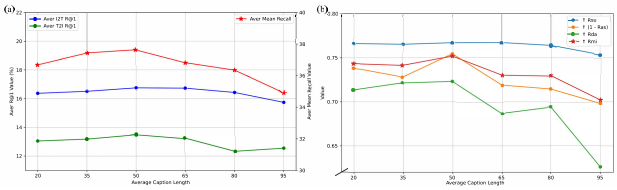


Fig. 14. Effectiveness of different caption lengths. (a) Image retrieval performance under different training dataset scales. (b) Semantic localization performance under different training dataset scales.

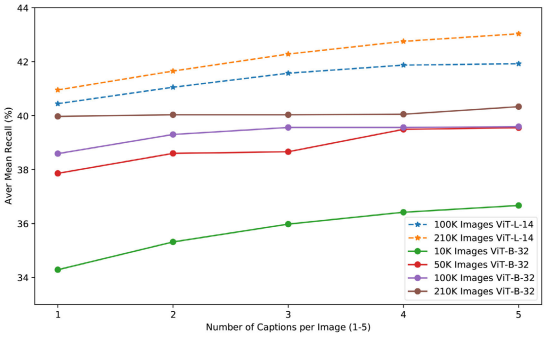


Fig. 15. Effectiveness of different caption numbers.

4) Effectiveness of Different Numbers of Captions per Image: In the second stage, we instructed the LLM to generate five slightly different captions per image to introduce diversity and randomly selected one caption for analysis to manage computational costs. To explore how training with multiple captions affects CLIP's performance, we tested four image scales—10K, 50K, 100K, and 210K—and provided each image with 1–5 captions. We then trained the CLIP model to observe performance changes with varying numbers of captions.

Fig. 15 shows that for smaller datasets (10K and 50K), the CLIP ViT-B-32 model's performance significantly improves as the number of captions increases, indicating that each caption adds unique information. However, for datasets larger than 100K, performance gains plateau despite more captions. This suggests that larger models are needed to fully utilize additional captions. Indeed, the CLIP ViT-L-14 model continued to show significant performance improvements at the 100K

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TABLE XV

COMPARISON OF TRAINING SUBDATASETS ON RSCTIR AND SELO TASKS. AVG MEANS AVERAGE

Data (Number)	Test	Avg Recall @1		Avg	Semantic Localization			
		I2T	T2I	Mean Recall	↑ Rsu	↓ Ras	↑ Rda	↑ Rmi
All (210K)	3ret	19.66	14.9	40.37	0.7635	0.2488	0.734	0.7518
CLS*8 (72K)	3ret	17.47	14.45	39.26	0.7665	0.2674	0.715	0.7417
DET-10 (82K)	3ret	16.13	13.48	36.11	0.7618	0.2957	0.6927	0.7244
Drone*4 (33K)	3ret	12.45	9.87	30.4	0.7475	0.2719	0.6522	0.7169
MSrgb*1 (22K)	3ret	12.34	10.56	31.26	0.7478	0.3044	0.6747	0.7113
CLIP-Baseline	3ret	8.02	7.67	24.32	0.7188	0.3006	0.6992	0.7071

TABLE XVI

COMPARISON OF CONSTRUCTION STRATEGIES IN RSCTIR AND SELO TASKS. BASELINE IS THE RESULT OF CLIP. R-M R MEANS RULE-MLLM RELAY GENERATION METHOD. THE BEST RESULT IS IN BOLD. AVG MEANS AVERAGE. GEN MEANS GENERATION. KOM MEANS KOSMOS-2. LVA MEANS LLAVA-1.6. MEC MEANS MECHANISM. PRT MEANS PROMPT

Model	Test	Avg Recall @1		Avg	Semantic Localization			
		I2T	T2I	Mean Recall	↑ Rsu	↓ Ras	↑ Rda	↑ Rmi
Baseline	3ret	8.02	7.67	24.32	0.7188	0.3006	0.6992	0.7071
R-M R	3ret	11.33	9.49	30.68	0.7125	0.3111	0.6833	0.6969
Lva-Gen	3ret	14.4	12.75	34.80	0.7609	0.2890	0.6718	0.7212
Kom-Gen	3ret	15.27	12.26	36.50	0.7414	0.2803	0.7063	0.7250
LLM-Prt1	3ret	18.75	13.82	38.46	0.7622	0.2765	0.7338	0.7415
LLM-Prt2	3ret	18.66	13.96	39.97	0.7594	0.2748	0.6996	0.7324
Fuse Mec	3ret	**19.66**	**14.90**	**40.37**	**0.7635**	**0.2488**	**0.7340**	**0.7518**

and 210K image scales, highlighting that increasing the model size is necessary to leverage larger datasets effectively.

5) Effectiveness of Different Subdatasets: To assess the impact of various subdatasets, we divided HQRS-IT-210K into Drone*4, DET-10, CLS*8, and MSrgb*1 based on their sources. We trained CLIP on each subdataset and evaluated the resulting models' performance to understand how different dataset types affect the model.

Table XV shows that CLS*8 and DET-10 significantly improve CLIP's RS retrieval capabilities, while Drone*4 and MSrgb*1 also enhance performance but to a lesser degree. This difference is mainly due to two factors: first, CLS*8 and DET-10 have 3 and 4 times as much data as Drone*4 and MSrgb*1, leading to better training results. Second, the images in CLS*8 and DET-10 are more similar to the test set, whereas Drone*4 and MSrgb*1 have lower distributional consistency with the test data. However, this inconsistency does not indicate lower quality of the Drone*4 and MSrgb*1 datasets. In the SeLo task, a similar trend is observed. Models trained with CLS*8 and DET-10 outperform those trained with Drone*4 and MSrgb*1 in terms of the Rsu metric, indicating better recognition of target regions.

6) Effectiveness of Different Construction Strategies: During dataset construction, we meticulously evaluated each strategy through independent experiments to validate its effectiveness. In the first stage, we generated image captions using three methods: **R-M Relay**, **LLaVA1.6-Generation**, and **Kosmos2-Generation**. In the second stage, we employed **LLM-Prompt1** and **LLM-Prompt2** to guide LLMs in summarizing the captions. At the end of the second stage, a fuse mechanism was used to combine the captions.

Table XVI demonstrates that performance consistently improved at each stage of caption generation, ultimately achieving the best results. This highlights the effectiveness of our dataset construction methodology. Additionally, we found that captions generated by Kosmos2 outperformed those from LLaVA1.6 after training CLIP. The primary reason is that LLaVA's captions were significantly longer (approximately 150 words) than Kosmos2's captions (approximately 44 words), leading to substantial text truncation during CLIP training and resulting in poorer performance.

7) Effectiveness of Different α Values: After the two caption generation stages, we found that mixing two types of LLM-generated captions increases the diversity of the training data and improves CLIP's performance. To identify the optimal mixing ratio, we conducted comparative experiments under different α values, where $\alpha = 1$ means that only Prompt 2 captions were used. The results show that an α of approximately 0.5 yields the best retrieval and SeLo performance, outperforming models trained with a single caption type.

This result suggests that the benefit does not come from any single caption style alone but from a balanced mixture of stylistically different yet complementary captions. From this perspective, α not only controls the proportion of the two prompt styles but also indirectly regulates the trade-off between semantic diversity and information concentration in the training captions.

E. Discussion on Applicability to Other Modalities

Although this work focuses on optical RS imagery, the core idea of MpGI is potentially extensible to other RS modalities, such as SAR and hyperspectral imagery. In essence, MpGI is a data-construction framework based on multisource semantic elicitation and subsequent semantic integration/distillation, rather than a modality-specific visual encoder design. Therefore, if suitable modality-specific caption sources, structured annotations, and foundation models are available, the overall generate–integrate paradigm could in principle be transferred beyond the optical domain.

However, such an extension is nontrivial and faces several modality-specific challenges. First, compared with optical imagery, SAR and hyperspectral data have fundamentally different imaging mechanisms and semantic characteristics. For SAR imagery, interpretation relies on microwave backscatter patterns and is often complicated by speckle noise, making semantic grounding less intuitive for general-purpose vision–language models. For hyperspectral imagery, many discriminative cues lie in fine-grained spectral signatures rather than directly perceptible appearance, which further increases the difficulty of natural-language description and cross-modal alignment [49].

Second, high-quality annotated datasets and mature modality-specific vision–language models remain much more limited for these modalities than for optical RS. Nevertheless, recent progress suggests that this direction is becoming increasingly feasible. For example, large-scale SAR image–text datasets and benchmarks have recently emerged, such as SARLANG-1M and SAR-TEXT, which provide new opportunities for multimodal learning and caption construction in SAR imagery [50], [51]. In addition, dedicated hyperspectral image captioning resources have also begun to appear,

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indicating the growing potential of vision–language learning beyond optical RS [52].

Overall, these developments suggest that the MpGI paradigm may be extended to nonoptical modalities in the future. However, additional modality-aware design and validation would still be necessary for both the multisource generation stage and the semantic integration stage. We therefore regard the extension of MpGI to SAR and hyperspectral imagery as a promising but still open direction for future work.

V. Conclusion

In this article, we proposed a **multiperspective semantic distillation paradigm** for RS image–text dataset construction and instantiated it as a two-stage framework, MpGI, by combining annotation-guided knowledge elicitation with semantic distillation and alignment-aware compression. Following this framework, we constructed the HQRS-IT-210K dataset, which provides high-quality and semantically rich image–text pairs for advancing RS vision-language learning. Our study suggests that, for RS image–text data construction, explicitly modeling complementary semantic views and distilling them into aligned textual representations is a more effective strategy than treating caption construction as a single-pass generation problem.

By fine-tuning CLIP and CoCa on this dataset, we obtained HQRS-CLIP and RS-CoCa. HQRS-CLIP is a robust discriminative VLFM that outperforms previous models across various RS applications while using substantially less training data, and RS-CoCa is a powerful generative model capable of producing captions comparable to or even better than those annotated by human experts. Extensive ablation experiments further validate the effectiveness of the proposed two-stage framework and provide useful insights into multiperspective semantic elicitation, integration, and distillation for future research.

References

[1] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 8748–8763.

[2] J. Yu, Z. Wang, V. Vasudevan, L. Yeung, M. Seyedhosseini, and Y. Wu, "CoCa: Contrastive captioners are image-text foundation models," 2022, *arXiv:2205.01917*.

[3] F. Liu et al., "RemoteCLIP: A vision language foundation model for remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[4] Z. Wang, R. Prabha, T. Huang, J. Wu, and R. Rajagopal, "SkyScript: A large and semantically diverse vision-language dataset for remote sensing," in *Proc. AAAI Conf. Artif. Intell.*, vol. 38, 2024, pp. 5805–5813.

[5] Z. Zhang, T. Zhao, Y. Guo, and J. Yin, "RS5M: A large scale vision-language dataset for remote sensing vision-language foundation model," 2023, *arXiv:2306.11300*.

[6] X. Li, C. Wen, Y. Hu, and N. Zhou, "RS-CLIP: Zero shot remote sensing scene classification via contrastive vision-language supervision," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 124, Nov. 2023, Art. no. 103497.

[7] L. D. Basso, "CLIP-RS: A cross-modal remote sensing image retrieval based on CLIP, a northern Virginia case study," M.S. thesis, Dept. Comput. Sci., Virginia Tech, Falls Church, VA, USA, 2022.

[8] Y. Bazi, L. Bashmal, M. M. Al Rahhal, R. Ricci, and F. Melgani, "RS-LLaVA: A large vision-language model for joint captioning and question answering in remote sensing imagery," *Remote Sens.*, vol. 16, no. 9, p. 1477, Apr. 2024.

[9] Y. Hu, J. Yuan, C. Wen, X. Lu, and X. Li, "RSGPT: A remote sensing vision language model and benchmark," 2023, *arXiv:2307.15266*.

[10] Y. Zhan, Z. Xiong, and Y. Yuan, "SkyEyeGPT: Unifying remote sensing vision-language tasks via instruction tuning with large language model," 2024, *arXiv:2401.09712*.

[11] S. Goyal, P. Maini, Z. C. Lipton, A. Raghunathan, and J. Z. Kolter, "Scaling laws for data filtering—Data curation cannot be compute agnostic," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 22702–22711.

[12] G.-S. Xia et al., "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.

[13] A. Abbas, K. Tirumala, D. Simig, S. Ganguli, and A. S. Morcos, "SemDeDup: Data-efficient learning at Web-scale through semantic deduplication," 2023, *arXiv:2303.09540*.

[14] S. Doveh et al., "Dense and aligned captions (DAC) promote compositional reasoning in VL models," in *Proc. Adv. Neural Inf. Process. Syst.*, 2023, pp. 76137–76150.

[15] J. Urbanek, F. Bordes, P. Astolfi, M. Williamson, V. Sharma, and A. Romero-Soriano, "A picture is worth more than 77 text tokens: Evaluating CLIP-style models on dense captions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 26690–26699.

[16] W. Zhou, S. Newsam, C. Li, and Z. Shao, "PatternNet: A benchmark dataset for performance evaluation of remote sensing image retrieval," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 197–209, Nov. 2018.

[17] X. Lu, B. Wang, X. Zheng, and X. Li, "Exploring models and data for remote sensing image caption generation," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2183–2195, Apr. 2018.

[18] Z. Yuan, Z. Xiong, L. Mou, and X. X. Zhu, "ChatEarthNet: A global-scale, high-quality image-text dataset for remote sensing," 2024, *arXiv:2402.11325*.

[19] Z. Lai et al., "VeCLIP: Improving CLIP training via visual-enriched captions," in *Proc. Eur. Conf. Comput. Vis.-ECCV*, 2025, pp. 111–127.

[20] B. Qu, X. Li, D. Tao, and X. Lu, "Deep semantic understanding of high resolution remote sensing image," in *Proc. Int. Conf. Comput., Inf. Telecommun. Syst. (CITS)*, Jul. 2016, pp. 1–5.

[21] Z. Yuan et al., "Exploring a fine-grained multiscale method for cross-modal remote sensing image retrieval," 2022, *arXiv:2204.09868*.

[22] Q. Cheng, H. Huang, Y. Xu, Y. Zhou, H. Li, and Z. Wang, "NWPU-captions dataset and MLCA-net for remote sensing image captioning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.

[23] M. Czerkawski and A. Francis, "From LAION-5B to LAION-EO: Filtering billions of images using anchor datasets for satellite image extraction," 2023, *arXiv:2309.15535*.

[24] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," 2018, *arXiv:1810.04805*.

[25] C. Raffel et al., "Exploring the limits of transfer learning with a unified text-to-text transformer," *J. Mach. Learn. Res.*, vol. 21, no. 140, pp. 1–67, 2020.

[26] M. Lewis et al., "BART: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension," 2019, *arXiv:1910.13461*.

[27] A. Radford, K. Narasimhan, T. Salimans, and I. Sutskever, "Improving language understanding by generative pre-training," OpenAI, San Francisco, CA, USA, 2018.

[28] A. Radford et al., "Language models are unsupervised multitask learners," *OpenAI Blog*, vol. 1, no. 8, p. 9, 2019.

[29] T. B. Brown et al., "Language models are few-shot learners," 2020, *arXiv:2005.14165*.

[30] J. Achiam et al., "GPT-4 technical report," 2023, *arXiv:2303.08774*.

[31] J. Li, D. Li, S. Savarese, and S. Hoi, "BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," in *Proc. Int. Conf. Mach. Learn.*, 2023, pp. 19730–19742.

[32] W. Dai et al., "InstructBLIP: Towards general-purpose vision-language models with instruction tuning," 2023, *arXiv:2305.06500*.

[33] S. Huang et al., "Language is not all you need: Aligning perception with language models," in *Proc. Adv. Neural Inf. Process. Syst.*, 2023, pp. 72096–72109.

[34] Z. Peng et al., "Kosmos-2: Grounding multimodal large language models to the world," 2023, *arXiv:2306.14824*.

[35] H. Liu, C. Li, Q. Wu, and Y. J. Lee, "Visual instruction tuning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2023, pp. 34892–34916.

[36] H. Liu, C. Li, Y. Li, and Y. J. Lee, "Improved baselines with visual instruction tuning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 26296–26306.

[37] H. Liu et al., "LLaVA-next: Improved reasoning, OCR, and world knowledge," Jan. 2024. [Online]. Available: https://llava-vl.github.io/blog/2024-01-30-llava-next/

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[38] J. Bai et al., "Qwen-VL: A frontier large vision-language model with versatile abilities," 2023, *arXiv:2308.12966*.
[39] C. Yang, Z. Li, and L. Zhang, "Bootstrapping interactive image–text alignment for remote sensing image captioning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5607512.
[40] W. Wang et al., "CogVLM: Visual expert for pretrained language models," 2023, *arXiv:2311.03079*.
[41] A. Young et al., "Yi: Open foundation models by 01.*AI*," 2024, *arXiv:2403.04652*.
[42] Z. Ji et al., "Survey of hallucination in natural language generation," *ACM Comput. Surveys*, vol. 55, no. 12, pp. 1–38, Dec. 2023.
[43] D. M. Chan, A. Myers, S. Vijayanarasimhan, D. A. Ross, and J. Canny, "IC3: Image captioning by committee consensus," 2023, *arXiv:2302.01328*.
[44] S. Teo, "How I won Singapore's GPT-4 prompt engineering competition," *Towards Data Sci.*, Dec. 2023. [Online]. Available: https://towardsdatascience.com/how-i-won-singaporesgpt-4-prompt-engineering-competition-34c195a93d41
[45] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," 2017, *arXiv:1711.05101*.
[46] A. Van Den Oord, Y. Li, and O. Vinyals, "Representation learning with contrastive predictive coding," 2018, *arXiv:1807.03748*.
[47] C. Qiu et al., "Few-shot remote sensing image scene classification: Recent advances, new baselines, and future trends," *ISPRS J. Photogramm. Remote Sens.*, vol. 209, pp. 368–382, Mar. 2024.
[48] J. L. Fleiss, "Measuring nominal scale agreement among many raters," *Psychol. Bull.*, vol. 76, no. 5, pp. 378–382, Nov. 1971.
[49] L. Pang, J. Yao, K. Li, J. Zhou, D. Meng, and X. Cao, "SPECIAL: Zero-shot hyperspectral image classification with CLIP," 2025, *arXiv:2501.16222*.
[50] Y. Wei et al., "SARLANG-1M: A benchmark for vision–language modeling in SAR image understanding," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026.
[51] Y. He et al., "SAR-TEXT: A large-scale SAR image-text dataset built with SAR-narrator and a progressive learning strategy for downstream tasks," 2025, *arXiv:2507.18743*.
[52] A. Das et al., "HyperCap: Hyperspectral land cover captioning dataset for vision language models," 2025, *arXiv:2505.12217*.



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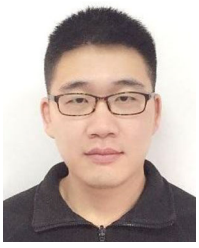
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